Bocconi University

Master of Science in Economic and Social Sciences

Household Wealth Inequality in Italy: Regional Differences and the Role of Asset Returns

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Academic Year:

2024–2025

Abstract

This thesis investigates the distribution of household wealth across Italian macro areas and the role of asset returns in shaping wealth inequality between 2006 and 2022. By combining SHIW microdata with National and Regional Accounts, and correcting for top-tail underrepresentation through Forbes rich lists and Pareto adjustments, the study reconstructs internally consistent wealth distributions for the North, Centre, and South & Islands. The findings show that, despite a general decline in real aggregate wealth, concentration at the top increased. By 2022, the Top 1% in the North held a wealth share about five times that of the Bottom 50%. Moreover, portfolio composition differs systematically across wealth groups: wealthier households hold a larger share of risky financial assets, while lower groups rely predominantly on housing. The results reveal that average portfolio returns increase along the wealth distribution, implying that differences in asset exposure contribute to persistent divergence in wealth accumulation. Finally, counterfactual simulations indicate that asset returns alone explain a substantial share of top wealth dynamics. For the Top 1%, returns on wealth account for approximately 85–87% of the observed variation in wealth growth in the North and 73–93% in the Centre, depending on the fixed portfolio specification.

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Introduction

How is household wealth distributed across Italian regions? And to what extent can asset returns explain the evolution of wealth and its concentration over time?

Historically, Italy has displayed persistent and pronounced territorial disparities since the unification period in the nineteenth century. These disparities are reflected not only in income, employment, and productivity but also in the accumulation and composition of household wealth. However, while regional income inequality has received attention in the literature, the regional dimension of wealth inequality remains comparatively underexplored.

Understanding how wealth is distributed geographically is particularly important in the Italian context for several reasons. For example, wealth taxation on housing is linked to local cadastral valuations and municipal tax rates. At the same time, the value of key wealth components, such as real assets, which represent a substantial share of Italian household portfolios, is strongly influenced by local public interventions and territorial characteristics. As a consequence, reconstructing regional wealth distributions is a necessary step to inform public policy debates on taxation, redistribution, and territorial cohesion.

Beyond spatial disparities, wealth evolution depends on multiple mechanisms. A large strand of literature highlights the roles of income inequality, intergenerational transfers, inter vivos gifts, socio-economic background, and institutional settings as key determinants of wealth concentration. In this study, the focus is placed on a specific but increasingly relevant channel: the role of asset returns. Wealth may grow not only through active saving decisions, but also through passive appreciation driven by capital gains and financial returns. If wealthier households hold portfolios that are structurally more exposed to high yield and riskier assets, differences in returns alone may

generate divergence in wealth levels and concentration.

This work contributes to the literature in three main ways. First, it reconstructs internally consistent wealth distributions for the North, Centre, and South & Islands by integrating survey microdata with National and Regional Accounts and correcting for top-tail underrepresentation. Second, it shows territorial heterogeneity in both wealth levels and inequality, showing that wealth concentration has increased particularly in the North, even as average real wealth has declined for the majority of the distribution. Third, it implements a counterfactual portfolio simulation to isolate the contribution of asset returns to the evolution of wealth. The results indicate that returns alone explain a substantial share of the variation in top wealth dynamics, especially in the North and in the Centre. This suggests that structural differences in exposure to high return assets play a central role in shaping wealth divergence.

By combining a regional perspective with an analysis of return-driven wealth dynamics, this study provides new evidence on the geography of wealth inequality in Italy and contributes to the broader debate on the mechanisms through which wealth concentration evolves over time.

The remainder of the work is organized as follows. Chapter 1 reviews the relevant literature on measuring and explaining wealth inequality. Chapter 2 describes the data sources and the methodological framework used to construct regional wealth distributions and to implement the simulation exercise. Chapter 3 presents the empirical findings. Chapter 4 discusses limitations and areas for future research. The final chapter concludes.

Chapter 1: Literature Review

This chapter reviews the main strands of literature relevant to the analysis of wealth inequality. It first discusses methodological approaches to measuring wealth distribution and territorial disaggregation. It then turns to the determinants of wealth inequality, with particular attention to the role of returns to wealth.

1.1 Measuring the Wealth Distribution

Research in economic inequality traces its origins to the seminal work of Kuznets (1955), who first proposed a long-run relationship between economic development and income inequality. In recent decades, this early framework has evolved into a much richer and empirically grounded line of research, largely driven by the contributions of Piketty (2014) and the contributions produced within the *World Inequality Database*[1]. Recent studies have profoundly reshaped our understanding of inequality by uncovering trends in concentration and emphasizing the central role of institutional and policy choices. Within this broader research stream, scholars increasingly recognize that wealth is systematically more unequally distributed than income across virtually all economies (Balestra and Tonkin, 2018). However, despite this flourishing literature, research specifically focused on wealth inequality remains comparatively limited, in part due to data limitations and methodological challenges.

Some of the first systematic attempts to estimate household wealth distributions tried to infer wealth from administrative sources such as tax records (Kopczuk and Saez, 2004). The pioneering work of Lampman (1962) for the United States and the subsequent refinements by Atkinson and Harrison (1978) for the United Kingdom are

[1]The World Inequality Database (WID) is an open access platform providing harmonized long-run series on income and wealth distribution across countries, based on national accounts, fiscal, survey, and financial data. It is maintained by the World Inequality Lab (Paris School of Economics).

early examples of innovative methodological approaches. With the expansion of survey microdata in the late twentieth century, researchers increasingly employed household surveys, most notably the U.S. Survey of Consumer Finances (SCF), analyzed by Wolff (1987), to study the distributional structure of wealth, portfolio composition, and the evolution of wealth concentration. Survey data provide broad population coverage and rich socio-demographic information, including poorer individuals not captured in tax registers. However, they also suffer from well-known drawbacks: underreporting, unit non-response, and underrepresentation of the top tail, leading to downward biases in measured wealth inequality (OECD, 2013). For these reasons, modern methodologies combine surveys with parametric Pareto tail adjustments, and, where available, external sources such as rich lists to correct the upper end of the distribution (Vermeulen, 2018). In parallel, increasing attention has been devoted to the use of National Accounts aggregates as a benchmark for distributional analysis, given their institutional credibility and transparency. More recently, the work of Saez and Zucman (2016), Piketty et al. (2018) and the DINA project[2] (Chancel et al., 2025) has introduced integrated frameworks that combine survey data, tax records, and national accounts to produce internally consistent wealth distributions. Such reconciliation is based on the premise that macroeconomic statistics provide the most reliable measures of wealth, while micro data are indispensable for distributional analysis; hence, neither source alone is sufficient for assessing wealth inequality[3].

In Italy, systematic research on wealth inequality has developed more gradually, largely constrained by data limitations and by the absence of administrative registers comparable to those available in other countries. The Bank of Italy's Survey on House-

[2]The DINA (Distributional National Accounts) project, coordinated by the World Inequality Lab, aims to reconcile micro data and national accounts in order to produce consistent distributional series of income and wealth.

[3]On the institutional side, this objective has motivated several coordinated initiatives aimed at standardizing micro-macro reconciliation practices in distributional analysis, including the *ECB Expert Group on Linking Micro and Macro Data* and the *OECD Expert Group on the Distribution of Household Wealth*. These efforts are closely aligned with Recommendation 9 of the G20 *Data Gaps Initiative*, which calls for the development of distributional information consistent with sectoral national accounts.

hold Income and Wealth (SHIW) has served as the primary source for studying household wealth, with the first analyses of the 1990s documenting the portfolio structure of Italian households and the overall level of wealth inequality (Brandolini et al., 2004). Subsequent contributions extended this perspective backward in time, exploiting the earliest survey waves to reconstruct wealth dynamics from the 1970s onward (Cannari and D'Alessio, 2018). More recent research has highlighted the limitations of survey data in capturing the top of the distribution and has therefore increasingly relied on alternative data sources. For instance, by integrating inheritance tax data, Acciari et al. (2024) show that top wealth shares in Italy are substantially higher than previously estimated. These studies reveal both the magnitude of wealth inequality in Italy and the methodological challenges inherent in its measurement, which motivate the methodological attention adopted in this work.

Recent years have seen a growing interest in understanding how wealth is distributed not only within a single nation, but also across space, giving rise to research on regional wealth inequality. This line of research highlights that wealth accumulation can have different territorial patterns, reflecting structural and institutional differences. In the United States, Suss et al. (2024) construct the GEOWEALTH-US database[4] to document spatial wealth inequality at subnational level from 1960 to 2000. A parallel effort for Europe is represented by the Spain Wealth Atlas, where Marrero et al. (2025) provide harmonized regional wealth estimates, revealing persistent and widening gaps across Spanish provinces and autonomous communities. For Italy, long-run historical work by Alfani and Ammannati (2017) on the Florentine State, by Alfani (2015) on Northwestern Italy, and by Alfani and Sardone (2025) on the Kingdom of Naples shows that territorial inequality has deep historical roots and has often been shaped by institutional, demographic, and economic shocks. For the contemporary period, re-

[4]The GEOWEALTH project develops a spatial database of wealth inequality in the United States from the 1960s onward, combining public-use microdata with machine learning and spatial computation techniques. The project is a collaboration between researchers at the University of Toronto and Arizona State University.

search has relied on reconstruction exercises by the Bank of Italy. First, Cannari et al. (2002) regionalized national aggregates using demographic and economic indicators. Alternatively, Cannari and D'Alessio (2003) exploited the SHIW household survey by pooling several waves together to increase sample size and obtain regionally representative estimates of household wealth. Nevertheless, the availability of harmonized and macro-consistent regional wealth distributions remains limited, leaving significant gaps in our understanding of how wealth inequality varies across Italian macro areas.

1.2 The Determinants of Wealth Inequality

Beyond measurement challenges, a substantial body of research has also explored the main drivers of wealth inequality. Many studies emphasize the role of income inequality as a determinant of diverging wealth accumulation patterns (Berman et al., 2016; Benhabib et al., 2017). However, they also show that income differences play a marginal role in accounting for the sharp divergence at the top of the wealth distribution. Household characteristics, such as educational attainment, financial literacy, employment stability, and demographic composition, further shape wealth outcomes (Bartscher et al., 2020; Antunes and Ercolani, 2020; De Beckker et al., 2025). Intergenerational factors can also play a central role: inheritances and inter vivos transfers contribute significantly to wealth concentration by reinforcing advantages across generations (Wolff, 2002; Adermon et al., 2018). Moreover, the influence of public policies, including taxation, monetary policy and pension reforms has been widely explored in relation to wealth inequality dynamics (Benhabib et al., 2011; Piketty and Saez, 2013; Andersen et al., 2024; Časta, 2025).

While these factors are important, increasing attention has been devoted to the role of returns on wealth as a central driver of wealth concentration, particularly at the top. Households differ systematically in their exposure to high-yield asset classes, and this

unequal access can generate persistent divergence in wealth trajectories. Empirical evidence highlights the importance of accounting for portfolio composition and asset specific returns when analyzing wealth inequality (Xavier, 2020; Pfeffer and Waitkus, 2021). Recent administrative evidence shows that returns to wealth are not only heterogeneous across asset classes but also systematically higher, more persistent, and strongly correlated with wealth at the top of the distribution, implying that differential returns can account for a substantial share of the rise in wealth concentration (Bach et al., 2020; Fagereng et al., 2020). These findings suggest that wealth inequality may rise because wealthier households are structurally positioned in portfolios that yield higher returns, making it essential to explicitly account for asset specific returns and portfolio allocation.

The existing literature highlights both the relevance and the methodological challenges of measuring wealth inequality, particularly when aiming for geographical disaggregation. To the best of my knowledge, to date, no study has constructed macro area wealth distributions for Italy using a fully integrated approach that combines survey microdata, external top tail information, and macroeconomic aggregates. This work contributes to filling this gap by deriving wealth distributions for the North, Centre, and South & Islands as if they were distinct economies, thereby allowing for spatially disaggregated and internally consistent analyses of wealth concentration.

In addition, this study connects the regional dimension of inequality with the broader literature on heterogeneous returns to wealth, and is among the first studies to provide a systematic assessment of how asset specific returns and portfolio composition contribute to the evolution of wealth inequality in the Italian context. Together, these contributions offer a more comprehensive mapping of wealth inequality in Italy and a richer empirical basis for understanding both its geography and its underlying drivers, with relevant implications for policies aimed at addressing persistent regional divides.

Chapter 2: Data and Methodology

This chapter presents the data sources and the methodological framework underlying the empirical analysis. It first introduces the SHIW microdata and the National and Regional Accounts used to construct consistent wealth distribution, and it then describes the adjustments implemented to reconcile differences across sources. The chapter concludes by outlining the construction of asset returns and the counterfactual portfolio simulation designed to isolate the role of returns in shaping wealth dynamics.

2.1 The Survey on Household Income and Wealth

The primary available source of microdata on household wealth in Italy is the Survey on Household Income and Wealth (SHIW), conducted by the Bank of Italy (Banca d'Italia, 2024b). The survey collects detailed information on income, wealth, and savings behavior of Italian households since the 1960s. In its most recent editions, the SHIW covers approximately 8000 households per year across more than 300 municipalities.

In order to maintain time comparability with other sources of data employed in this study, the analysis focuses on SHIW waves from 2006 to 2022, covering a total of 8 editions.[1] The unit of analysis is the household.

As already discussed in the literature review, survey data are subject to limitations, particularly in their ability to adequately capture the upper tail of the wealth distribution. One major concern is non-response bias, which may distort estimates if participation in the survey is systematically correlated with socio-economic characteristics. According to the Bank of Italy's methodological documentation,[2] non-response bias in the SHIW is positively correlated with both income and the education level of the household head,

[1] The survey data are available for 2006, 2008, 2010, 2012, 2014, 2016, 2020, and 2022.

[2] For a detailed discussion of the methodology, see the methodological notes published by the Bank of Italy.

potentially leading to an under representation of wealthier households. This issue is partially mitigated by the weighting scheme adopted in the SHIW, as described by Faiella and Gambacorta (2007). The survey weights correct for differential participation probabilities and are calibrated to reproduce known population characteristics, including sex, age group, geographical area, and municipality size.

Moreover, the SHIW is made representative at the macro area level through post-stratification procedures. For this reason, and given the sampling variability observed at finer levels of disaggregation, the analysis is carried out at the macro area level (North, Centre and South & Islands), where the survey maintains its representativeness. This aggregation strategy is consistent with previous research on income inequality, such as Guzzardi et al. (2024), which also relies on macro regional comparisons to ensure robust and comparable estimates across areas.

However, the weighting framework cannot fully address another important limitation of survey data, namely under reporting. The provision of incomplete or inaccurate information may vary across the distribution, across regions, and by asset type. For example, D'Aurizio et al. (2006) documents that financial assets are more likely to be under reported by households residing in the North or belonging to the upper tail of the wealth distribution. To mitigate these residual discrepancies, the survey microdata are reconciled with macroeconomic aggregates, as discussed in the following section.

2.2 Macro Aggregates

To ensure consistency between microdata and aggregate wealth totals, the analysis integrates SHIW data with National and Regional Accounts, which are widely considered the most reliable aggregate source for wealth measurement (Piketty et al., 2018).

2.2.1 National Accounts

The Italian National accounts data used in this study come from the *Conti patrimoniali*, a joint publication by the Bank of Italy and the Italian National Institute of Statistics (ISTAT) (Banca d'Italia, 2024a). These data provide a comprehensive overview of the balance sheets of Italy's institutional sectors, with particular attention to the household sector between 2005 and 2023.

The National Accounts follow a harmonized framework for measuring net wealth, consistent with the *European System of Accounts (ESA 2010)* and the *System of National Accounts (SNA 2008)* (Istat and Banca d'Italia, 2025; European Commission et al., 2009). Total net wealth is defined as the sum of real and financial assets minus financial liabilities. The real components include *dwellings*, *non-residential buildings*, *plants and machinery*, *intellectual property products*, *biological resources*, *inventories*, and *land* together with *land improvements*. Financial assets comprise *currency and deposits*, *securities, shares and equity holdings*, *mutual fund units, pension and insurance reserves*. Financial liabilities represent the negative components of wealth and consist mainly of *mortgages* and *personal loans*.

Real assets are estimated by ISTAT, using market prices and indirect valuation methods, while financial assets and liabilities are drawn from the Financial Accounts of the Bank of Italy, valued at current market prices. A distinctive feature of the Italian approach is that the value of dwellings and non-residential buildings is reported including the value of the underlying land, diverging from the ESA 2010 classification. Moreover, some non-produced assets, such as monuments, works of art, and natural resources other than land, are not yet valued. These conceptual differences must be taken into account in international comparisons and when bridging the gap between micro and macro sources.

The household sector in these accounts includes both consuming and producing

households as well as non-profit institutions serving households (NPISHs), that is, private entities that produce non-market goods and services for households, such as trade unions, sports associations, political parties, and similar organizations. According to the System of National Accounts, NPISHs are institutional units that are separate from households and operate on a non-profit basis for the benefit of households as a community (European Commission et al., 2009). This definition implies a broader coverage than that of the Survey on Household Income and Wealth (SHIW). As a result, direct comparison between the two sources requires careful adjustments.

Moreover, according to the System of National Accounts, unincorporated household enterprises that are not legally distinct from their owners are included in the household sector, whereas those that behave as autonomous entities are treated as quasi-corporations and thus reclassified to the corporate sector. In the Italian implementation of these principles, producing households are defined as those carrying out an unincorporated economic activity employing no more than five persons (Banca d'Italia, 2025). In contrast, in the SHIW, the household sector covers all households engaged in productive activities, regardless of their legal form or number of employees. This divergence may affect the measurement of business wealth and must therefore be taken into account.

The following sections describe in detail how these discrepancies are addressed to ensure internal consistency in the reconciliation procedure.

2.2.2 Regional Accounts

Although National Accounts provide the most reliable benchmark for aggregate household wealth, they do not offer a direct territorial breakdown at the macro area level. Relying exclusively on national totals would therefore require imposing arbitrary regional shares. To avoid this, the analysis incorporates Regional Accounts, which provide geographically disaggregated information and allow wealth aggregates to be allocated

across macro areas using observed territorial proportions.

Regional Accounts in Italy are derived from separate publications for each region, jointly based on data from the Bank of Italy and ISTAT.[3] The underlying wealth concept is broadly consistent with that used in the National Accounts, although components are more aggregated. Real assets are divided into *dwellings* and *other real assets*, while financial assets are grouped into *(i) currency and deposits; (ii) debt securities, shares, holdings, mutual funds, and loans to cooperatives;* and *(iii) other financial assets*. Financial liabilities are classified into *loans* and *other financial liabilities*.[4]

Data for 2012-2022 were retrieved from the 2025 regional publications for all regions, while earlier years were collected from previous releases. This introduces a limitation, as older publications are subject to larger revisions and tend to be less accurate,[5] particularly for specific asset categories (Albareto et al., 2008). To mitigate this issue, the 2025 National Accounts were adopted as the benchmark, and regional aggregates were re-scaled so that their macro area totals matched national values. As expected, recent years aligned almost perfectly with the National Accounts, while earlier data required more substantial adjustments. The magnitude of these adjustments is reported in Appendix A.1, which presents the correction factors applied by asset or liability and year.

Finally, internal consistency was verified by comparing macro totals with the sum of their sub-components. The resulting residual discrepancies, reported in Appendix A.2, are negligible (generally below 1%), confirming that rounding differences are the main sources of mismatch. To prevent inconsistencies from affecting the subsequent analysis, internal accounting coherence at the micro level was imposed. Specifically, macro

[3]Regional data are available in the *Economie Regionali* series published by the Bank of Italy, specifically in the statistical appendix under the table “La Ricchezza delle famiglie”. Publications can be accessed through the Bank of Italy website.

[4]For a detailed description of the methodology used to construct regional wealth aggregates, see *Rapporti annuali regionali – Note metodologiche*

[5]The only missing values were real assets and financial liabilities 2006 data for Trento and Bolzano provinces, which were interpolated linearly.

aggregates were reconstructed as the sum of their underlying components, rather than relying mechanically on reported totals, thereby guaranteeing identity between child and parent items.

2.3 Adjustments

Given the conceptual and methodological discrepancies between the SHIW and the National and Regional Accounts, specific adjustments are introduced to ensure consistency before reconciliation.

2.3.1 Conceptual differences

Price adjustments

First, to enhance time comparability and to account for inflation, all values both from the SHIW and the National and Regional Accounts have been deflated using the Index of Monetary Revaluation produced by ISTAT.[6] The reference year will be 2022, so that every level value will be expressed in 2022 constant prices.

Wealth components

Then, to reconcile the definition of wealth components between the micro and macro estimates, a detailed alignment of items was required. The detailed mapping of each wealth component and its inclusion in either the SHIW or the National and Regional Accounts is presented in the Appendix A.3.

With respect to real assets, the National and Regional Accounts do not explicitly report the value of valuables, defined as produced assets such as artworks, jewelry, and other collectibles that are expected to appreciate over time, mainly due to measurement

[6]The monetary revaluation index is provided directly within the SHIW dataset.

and valuation difficulties (Istat and Banca d'Italia, 2025). Consequently, this component was deducted from SHIW real and net wealth to ensure comparability. This adjustment has a negligible impact on aggregate estimates, as valuables account for only about 2% of total non-financial assets.

Beyond this adjustment, an additional reclassification was required to harmonize the treatment of business wealth between the SHIW and the macroeconomic sources, as also noted in the literature (Baffigi et al., 2016). Following the methodology outlined in Bonci et al. (2005) and Banca d'Italia (2007), the business wealth of producing households with more than five employees, classified as quasi-corporations in the National Accounts, was removed from household wealth. This treatment reflects the fact that quasi-corporations belong to the corporate sector rather than the household sector in the National Accounts. Conversely, the business wealth of non-incorporated households producers (with up to five employees) was retained among real assets. On the other hand, shares in incorporated firms (such as Srl or SpA)[7] were moved under financial assets.

Focusing on financial assets and liabilities, government bonds, which are classified as a distinct category in the SHIW, were incorporated into the broader aggregate *Other financial instruments* to align with the Regional Accounts classification. Moreover, both credits and debts towards other households, for which no counterpart exists in the Financial Accounts, were excluded from SHIW household wealth. From the perspective of the Regional Accounts, since the disaggregation of insurance and social security reserves cannot be consistently reconstructed within the SHIW classification, the corresponding Regional Accounts item, called *Other financial assets*, was incorporated into the broader category of Other financial instruments for the purpose of comparison. However, for the empirical analysis, this aggregate is subsequently decomposed after

[7] In Italy, a Società a responsabilità limitata (S.r.l.) and a Società per azioni (S.p.A.) are limited liability companies broadly comparable to private limited companies (Ltd) and public limited companies (PLC) in the UK system, respectively.

reconciliation using SHIW micro-level information, distinguishing between bonds, public equity, and private equity. Table 2.1 presents a complete overview of the variables included in the wealth definition analyzed in this work.

Table 2.1: Wealth components

Variable	Description
Real assets / Non-financial assets (a)	
Dwellings	
Other real assets	Business wealth, non-residential buildings, land and land improvements, machinery, equipment and other structures
Financial assets (b)	
Currency and deposits	
Other financial instruments	Debt securities, shares, holdings, mutual funds, loans to cooperatives, insurance and social security reserves, commercial credits, and other accounts receivable
Financial liabilities (c)	
Loans	
Other financial liabilities	Commercial debt and other accounts payable
Net wealth (a + b - c)	

Non-profit institutions serving households (NPISH)

Further adjustments are applied to harmonize the definition of the household sector across data sources, ensuring consistency in the unit of analysis. While Bonci et al. (2005) estimated the NPISHs wealth shares as 2% for financial assets and 3% for financial liabilities based on computations for the period 1998-2002, I updated their approach using more recent data from *Eurostat* and the *World Inequality Database*.[8] These data sources provide separate series for households and NPISHs, allowing the computation of the NPISH share in total wealth, financial assets, and debts. Since no direct information is available for real assets, their share is estimated residually. For Italy, NPISHs account on average for about 5.7% of real assets, 1.2% of total financial assets, 1.1%

[8]Data used in this section are drawn from the Eurostat database and from the World Inequality Database, maintained by the World Inequality Lab.

of financial liabilities, and 4.1% of total net wealth over the analyzed period.[9] These estimated yearly shares are subtracted from the adjusted Regional Accounts aggregates. For simplicity, NPISH shares are assumed to be evenly distributed across macro areas. The detailed yearly series of NPISH weights used for the adjustment are reported in Appendix A.4.

Dwellings and land underlying dwellings

The estimation of dwellings in the National Accounts, which includes the value of land underlying buildings, should not pose consistency issues when compared with survey data. Although no publicly available SHIW documentation explicitly states that the variable for dwellings includes the value of the land on which the building stands,[10] the formulation of the question, which refers to the estimated market value of the entire property strongly suggests that the reported value should be interpreted as inclusive of the land, as also noticed by Santoro and Tartamella (2021).

2.3.2 Top Tail Adjustment

Since the SHIW does not oversample wealthier households, the upper tail of the distribution is systematically underrepresented, leading to a substantial downward bias in the measurement of top wealth shares. As shown by Vermeulen (2018), Italy, together with the Netherlands is one of the few advanced economies whose wealth survey lacks any top-tail oversampling. Information from rich lists, such as the *Forbes World's Billionaires List*,[11] can therefore be useful in mitigating this limitation. In the Italian case,

[9]The updated estimates are broadly consistent with the early figures reported by Bonci et al. (2005), confirming that NPISHs represent a very limited share of household-sector wealth.

[10]The relevant question in the SHIW questionnaire asks respondents to estimate the current market value of the dwelling in which they live, assuming it were vacant and including annexed spaces (such as cellars, garages, and attics), and referring to the entire property rather than the household's ownership share. For further details, see the SHIW questionnaires available from the Bank of Italy.

[11]The list is an annual ranking that reports wealth estimates for the world's richest individuals based on public information. Although not an official statistical source, it is widely used in the empirical literature to approximate the upper tail of the wealth distribution.

Table 2.2: Billionaires by macro-area and year

	North			Centre		
Year	# of Billionaires	Average Wealth (2022 Constant)	Total Wealth (2022 Constant)	# of Billionaires	Average Wealth (2022 Constant)	Total Wealth (2022 Constant)
2006	11	4.00	44.02	1	2.76	2.76
2008	10	3.50	34.95	1	2.15	2.15
2010	10	3.50	35.03	1	1.36	1.36
2012	12	3.67	44.09	2	1.02	2.04
2014	25	3.40	85.02	6	1.25	7.51
2016	36	2.65	95.22	4	3.58	14.31
2020	26	2.56	66.64	5	2.37	11.87
2022	43	2.96	127.30	5	2.79	13.96

Note: Total and average wealth are expressed in billions of 2022 constant euros. Billionaires are counted based on city of residence.

Forbes data[12] reveal an evident geographical concentration of billionaires. The disaggregation by macro area is based on the publicly reported primary residence of each billionaire. As documented in Table 2.2, the North consistently hosts the majority of ultra rich residents, and their aggregate wealth has grown sharply over time, reaching about €127 billion euros in 2022. The Centre records only a few observations, with a maximum of 6 billionaires in 2014, while no billionaire has ever been reported as residing in the South & Islands over the period considered. This evidence further supports the use of external data to model the top tail, since billionaire wealth is not a marginal phenomenon.

To correct for the missing upper tail, I follow the five step framework of Bach et al. (2019). First, the richest SHIW households are combined with Forbes entries to form an enriched upper tail sample. Second, conditional on exceeding a data-driven wealth threshold, a weighted Pareto model is estimated separately for each macro area and year. In the North and Centre, where billionaire information is available, the Pareto shape parameter is estimated using both survey observations and rich list data. In the South & Islands, the Pareto parameter is estimated exclusively from SHIW data, with particular attention devoted to smoothing the tail exponent to avoid implausible behavior.

[12]For the period up to 2014, billionaire data are taken from Acciari et al. (2024). The 2016 list was obtained from the public archive maintained by Areppim, while the 2020 and 2022 lists were retrieved from Wikipedia. All lists were cross-checked, cleaned, and harmonized for consistency.

Third, synthetic wealth values are simulated from the estimated distributions. Fourth, these simulated values replace the original SHIW observations above the threshold, leaving the remainder of the distribution unchanged. Finally, the reconstructed upper tail is merged with the empirical bottom tail to obtain the complete wealth distribution for each macro area. The adjustment parameters for selected years are reported in Appendix A.5, while the impact of the adjustment on top 1% wealth shares is shown in Appendix A.6. On average over the period 2006-2022, the Pareto adjustment increases the top 1% share by about 2.5% in the North, 1.9% in the Centre, whereas the effect is negligible in the South & Islands.

A further adjustment concerns the attribution of portfolio components to synthetic top-tail households, which report only total net wealth and no information on the composition of assets and liabilities. To address this issue, I follow a simplified and transparent approach similar to that used by Cantarella et al. (2021). For each macro area and year, portfolio shares relative to net wealth are computed from the richest fifty survey households. These shares are then applied proportionally to the synthetic wealth values, leaving all survey observations unchanged. This ensures that the reconstructed top tail is internally consistent and assigns portfolio structures to the reconstructed top tail.

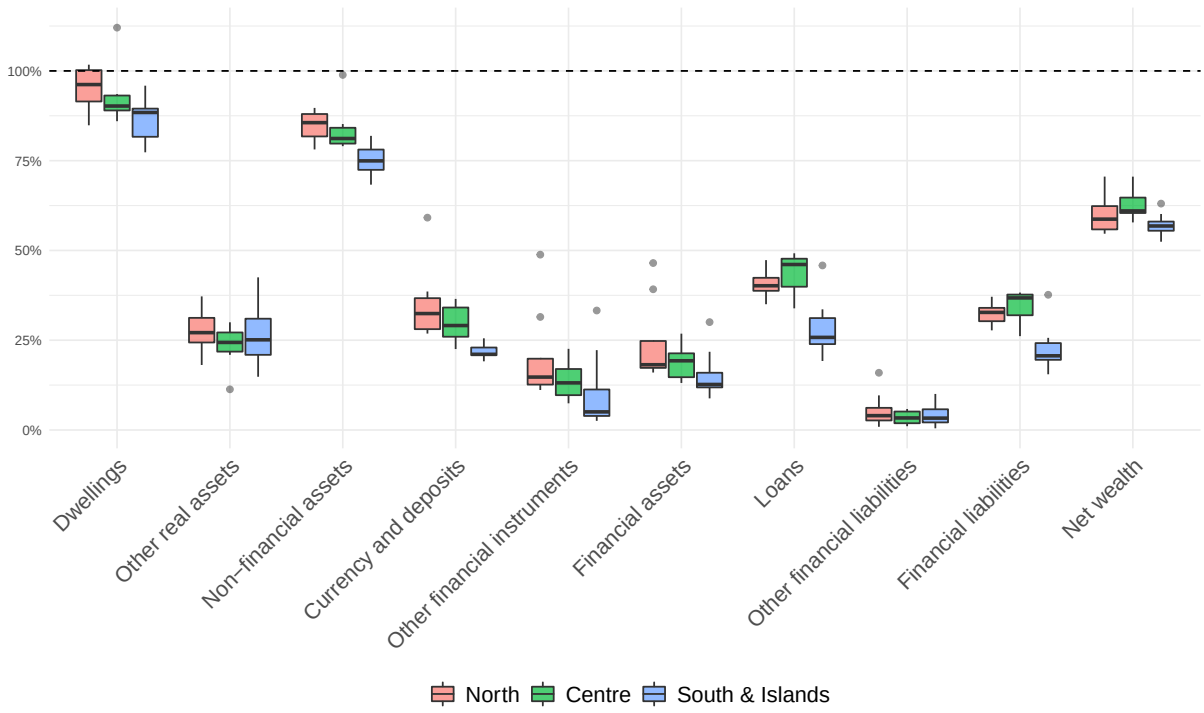
2.3.3 Micro-Macro Reconciliation

After correcting the upper tail using external information, a further adjustment is required to address the under-reporting that characterize survey data, particularly at the top of the distribution (Zwijnenburg, 2022). This involves reconciling the adjusted micro distribution with the corresponding macroeconomic aggregates.

Figure 2.1 assesses the alignment between the SHIW microdata, after the Pareto adjustment, and the Regional Accounts components across macro areas. The boxplots pool all available survey years and display the distribution of coverage ratios, de-

fined as the ratio between micro-aggregated SHIW components and the corresponding Regional Account aggregates. Consistent with previous evidence, the coverage of *Dwellings* is the highest among all items, closely matching the macroeconomic statistics (Baffigi et al., 2016). By contrast, financial assets and liabilities exhibit more pronounced gaps, reflecting well-known measurement challenges in household surveys (D'Alessio and Faiella, 2002). *Net wealth* shows a relatively higher coverage, particularly in the North and the Centre, where the median ratio is around 60%, although a sizable gap still remains.

Figure 2.1: Coverage ratios of SHIW relative to Regional Accounts



Note: Coverage ratios are the ratio of micro data to the corresponding macro totals. Boxplots pool all available survey years for each macro area. Values below 100% indicate under-coverage of SHIW relative to Regional Accounts.

To perform the reconciliation, the *"Proportional Adjustment"* methodology proposed by D'Alessio and Neri (2015) is adopted, as it represents the simplest and least arbitrary procedure among those discussed in their study. The approach consists of rescaling each wealth component in the microdata by a correction factor equal to the ratio be-

tween its aggregate value in the Regional Accounts and the corresponding weighted SHIW total, for each macro area and year. This procedure ensures exact consistency between micro-level aggregates and the totals reported in the Regional Accounts.

2.4 Counterfactual Portfolio Simulation

This section describes the data and empirical strategy designed to isolate the contribution of asset returns from other mechanisms of wealth accumulation.

The approach relies on a counterfactual simulation exercise in which observed asset-specific returns are applied to fixed portfolio structures. Holding portfolio shares constant allows the simulation to isolate the effect of differential asset returns from changes in portfolio reallocation. In practice, returns are allowed to vary over time, while the composition of assets remains unchanged. This strategy is conceptually similar to that adopted by Cannari et al. (2008), who evaluate the contribution of capital gains to wealth inequality by fixing portfolio weights at a base year. By comparing the counterfactual wealth trajectories generated by the simulation with the observed evolution of wealth, the exercise provides a benchmark for evaluating how much of the change in wealth levels and concentration can be attributed to returns alone.

Formally, for each macro area a, wealth group g, year t, and asset j net wealth evolves according to the following law of motion:

$$NW_{a,g,t} = \left(1 + \sum_j s^{(0)}_{a,g,j}\, r_{a,j,t}\right) W_{a,g,t-1} - D_{a,g,t}, \tag{2.1}$$

where $NW_{a,g,t}$ denotes net wealth, $W_{a,g,t-1}$ denotes gross wealth of the previous period, $s^{(0)}_{a,g,j}$ represents the fixed share of asset j in the portfolio of wealth group g in macro area a at the reference year (base period), and $r_{a,j,t}$ denotes the observed real return on asset j in macro area a at time t. $D_{a,g,t}$ denotes the level of financial liabilities in

period t, which is observed in the data and treated as exogenous to the simulation.

To implement the simulation, the analysis combines information on household portfolio composition derived from the SHIW with external data on asset returns. Returns are constructed for the main asset categories held by households and are expressed in real terms, consistently with Fagereng et al. (2020), in order to ensure coherence with the real wealth measures used throughout the study. The simulation is performed separately for each macro area, treating the North, Centre, and South & Islands as distinct wealth distributions.

2.4.1 Construction of Asset Returns

In constructing asset returns, the analysis draws inspiration from the methodological framework developed by Fagereng et al. (2020). In line with their approach, returns are defined comprehensively as the sum of income flows and capital gains. This includes both realized and unrealized capital gains, as well as cash flows such as dividends, interest payments, and rental income.

Real assets constitute a central component of household portfolio and display an important geographical dimension, as their returns can be meaningfully decomposed at the macro area level. Consistently with Poterba (1992), the return on *Dwellings* is modeled as the sum of two components: capital gains, reflecting changes in housing prices over time, and rental yields, capturing the flow of income generated through rental contracts. The relative share of these two components is identified using micro-level information from the SHIW, which allows distinguishing between owner-occupied dwellings and properties rented out. Capital gains are constructed by integrating the ISTAT House Price Index (IPAB),[13] complemented by the ECB House Price Index[14] to extend the time coverage of the series. Due to the lack of disaggregated information

[13]House Price Index (IPAB) data are available from the ISTAT Data Explorer.
[14]Residential property price data for Italy are available from the ECB Statistical Data Warehouse.

in the early years, the regional breakdown is available only from 2011 onwards. For years prior to 2011, national price dynamics are used and assumed to be a proxy for all macro areas. Regarding the estimation of rental yields, the analysis combines national price-to-rent indicators with evidence on regional rental profitability. The price-to-rent ratio published by the OECD[15] is calibrated using estimates of net rental yields at the regional level reported by Guerrieri and Festa (2023).

The item *Other real assets* comprises a heterogeneous set of components such as business wealth, non-residential buildings, land, machinery, and equipments. Unlike *Dwellings*, estimating returns for this broader category is substantially more challenging due to the absence of comprehensive market price series and the intrinsic heterogeneity of the underlying assets. Capital gains are proxied using two sources. First, annual variations in quotations for productive real estate published yearly by the Observatory of the Real Estate Market (OMI) of the Revenue Agency.[16] Second, changes in agricultural land prices are obtained from the Council for Agricultural Research and Analysis of Agricultural Economics (CREA).[17] These two components are combined using their relative shares in the National Accounts. The income yield component is instead approximated using an income-based approach drawing from administrative tax data. Specifically, the stock is taken from the Regional Accounts balance sheets, while the corresponding flow of income is approximated using information from personal income tax declarations published by the Italian Ministry of Economy and Finance.[18] These data report income related to land, buildings, and other productive activities.

For financial assets, a territorial disaggregation of returns is not meaningful, as yields are primarily determined in integrated national and international markets rather than by local economic conditions. Returns on financial assets are therefore assumed

[15]Price-to-rent ratio data are available from the OECD Data Explorer.
[16]OMI data are taken from non-residential real estate relationships published every year.
[17]Land prices data are taken from the Land Value Database (BDVF).
[18]IRPEF income data are taken from tax statistics published by the Italian Ministry of Economy and Finance.

to be common across macro areas. Returns on *Deposits* are derived from interest rate statistics published by the Bank of Italy,[19] which provide consistent estimates of average remuneration on household deposits over time. The category of *Other financial instruments*, however, encompasses a highly heterogeneous set of assets with different risk and return profiles. To account for this heterogeneity, and consistently with the literature, financial instruments are further disaggregated into three broad asset classes: *Bonds*, *Public equity*, and *Private equity*.

In line with Smith et al. (2023), returns on *Bonds* are proxied using the benchmark yield on ten-year Italian government bonds (BTP), as published by the Bank of Italy.[20]

Moreover, returns on *Public equity* are constructed using a composite index that combines the MSCI World Index and the FTSE MIB Index, both obtained from Curvo.[21] In line with Fagereng et al. (2020), this composite measure is intended to capture both the international and the domestic components of Italian households' equity exposure through market return indices. For simplicity, the weights are held constant over time, assuming a portfolio composed of 15% global equities and 85% domestic equities. This calibration is guided by evidence from the SHIW and supported by analyses from the Bank of Italy,[22] which document a persistent home bias in the equity holdings of Italian investors.

Finally, returns on *Private equity* are constructed following a logic consistent with Bach et al. (2020), who emphasize the importance of accounting for unlisted business wealth. Given data limitations for Italy, accounting information on firm performance is used as a proxy. Specifically, the return is approximated by the average return on equity (ROE) of medium-sized enterprises in Italy, drawing on data published by the *Area Studi*

[19] Deposit returns are available through the Bank of Italy Statistical Database.

[20] BTP returns are available through the Bank of Italy Statistical Database.

[21] Equity returns are derived from the performances of MSCI World Index and the FTSE MIB Index, retrieved from Curvo.

[22] Further evidence on households' portfolio composition can be found in the Bank of Italy's *Survey on Household Income and Wealth* reports, available at the Bank of Italy website.

Mediobanca.[23] These reports track the evolution of profitability indicators for a large and representative sample of Italian unlisted firms over time. Although this proxy does not fully capture valuation changes, it provides a consistent and economically meaningful measure of returns on privately held business wealth in the Italian context.

Table 2.3 reports the average annual real returns of each asset class together with their corresponding standard deviations. The evidence broadly confirms the standard risk–return hierarchy for financial assets, with safer assets yielding lower average returns than riskier ones, and highlights macro-area heterogeneity for real assets.

Table 2.3: Average annual real returns by asset class (%, 2006–2022)

Panel A: Real assets

Asset class	**North**	**Centre**	**South & Islands**
Dwellings capital gains	-1.25 (3.00)	-2.06 (3.72)	-1.13 (2.66)
Dwellings rental yield	2.59 (0.35)	2.23 (0.30)	2.07 (0.28)
Other real assets capital gain	-2.05 (1.84)	-2.45 (2.21)	-1.62 (2.16)
Other real assets income yield	2.83 (0.58)	3.30 (0.35)	2.29 (0.25)

Panel B: Financial assets

Asset class	**Italy**
Deposits	-0.88 (1.84)
Bonds	1.47 (1.89)
Public equity	2.73 (20.21)
Private equity	5.98 (3.52)

Note: Returns are expressed in real terms. Values report mean annual return with standard deviation in parentheses, computed over the period 2006–2022.

[23] Data on the performance of medium-sized Italian enterprises are retrieved from several publications by *Area Studi Mediobanca*, available at the Mediobanca Research Area website.

Chapter 3: Empirical Findings

This chapter presents the empirical findings of the work. It begins with descriptive evidence on wealth levels, inequality, and portfolio composition across macro areas and wealth groups. It then evaluates how asset returns contribute to the evolution of wealth and its concentration.

3.1 Descriptive Evidence

3.1.1 Wealth Levels and Territorial Trends

Italy is a country where wealth plays a central role in the economic system. This is reflected in a wealth-to-GDP ratio persistently around 5.8,[1] in line with the evidence documented by Piketty and Zucman (2014), which places Italy at the upper end of advanced economies in terms of wealth relative to national income. Despite this structural importance, real household wealth declined between 2006 and 2022, falling from approximately €11.5 trillion to €10.5 trillion (in 2022 prices). A substantial part of this contraction is linked to the cumulative rise in prices: over 2006–2022, the revaluation coefficient implies aggregate inflation of nearly 30%, so that even relatively stable nominal values translate into declining real wealth.

Macro area patterns broadly mirror the national decline but reveal important asymmetries. Considering median real wealth, shown in Appendix B.1, the North follows a downward trajectory, with a particularly marked drop around 2012. The Centre records the highest median levels in the early years, exceeding €320 thousand, but also experiences the sharpest contraction over the period. The South & Islands display persistently

[1] GDP data for Italy and the three macro areas are taken from ISTAT Regional Accounts, available through the ISTAT Data Explorer.

lower median wealth, although their decline is comparatively smoother.

These aggregate patterns, summarized in Appendix B.1, provide the necessary context for the distributional analysis that follows, highlighting both the erosion of real wealth over time and the persistence of substantial territorial disparities.

3.1.2 Inequality Indicators

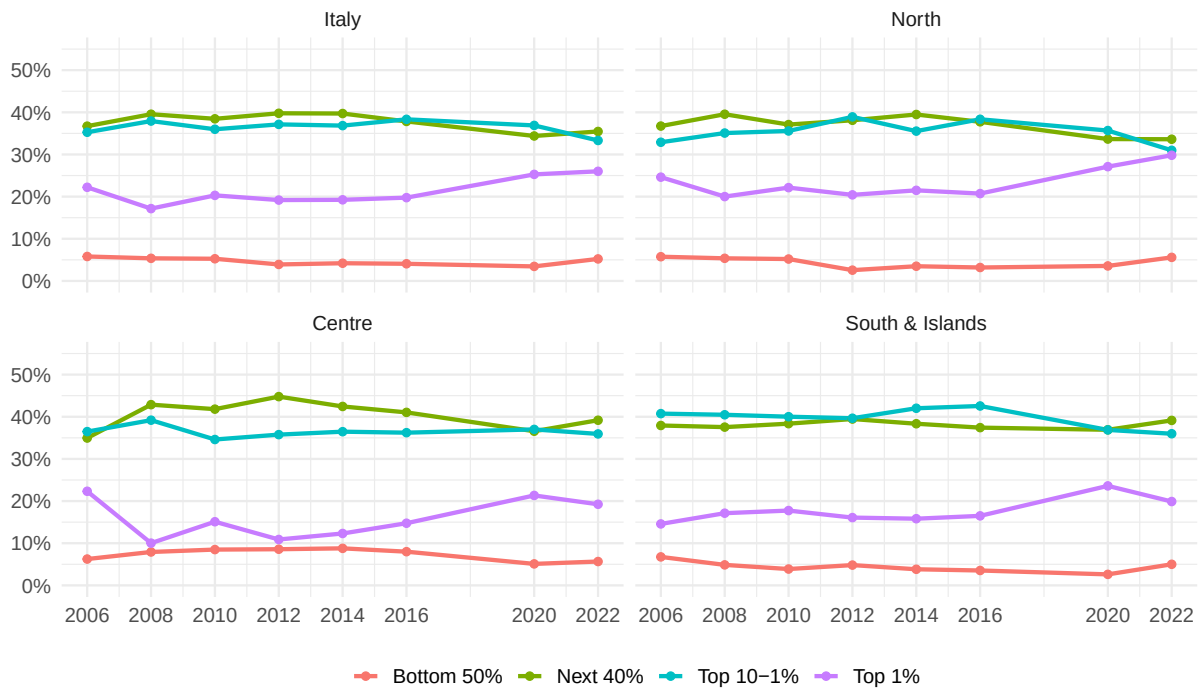
The analysis of wealth shares, shown in Figure 3.1, follows a standard approach in inequality studies, as it provides a direct representation of wealth concentration. At the national level, the shares held by the top 10-1% (households between the 90th and the 99th percentile) and the next 40% (households between the 50th and the 90th percentile) are relatively close and occasionally alternate in magnitude. Over time, however, the share of wealth controlled by the top 1% increased steadily reaching over 25% of total household wealth. The bottom 50%, by contrast, consistently owns around 5% of total wealth throughout the period, implying that a group fifty times larger than the top percentile holds less than one fifth of its wealth.

Macro area patterns reveal substantial heterogeneity. In the North, the top 1% follows a pronounced and persistent upward trajectory, capturing an increasing share of total wealth over time. This rise is accompanied by a gradual contraction in the shares of both top 10-1% and the next 40%, signaling a weakening of the upper middle class. By 2022, the top 1% reaches a wealth share comparable to that of the top 10-1% and only slightly below that of the next 40%, highlighting the strength of concentration dynamics in the North. In the Centre, the next 40% generally maintains a higher share than the top 10-1%, while the top 1% exhibits noticeable fluctuations. In the early years, the share of the top 1% approaches that of the bottom 50%, highlighting a relatively compressed structure of the distribution. Nevertheless, even in the Centre an upward trend in the top 1% share emerges toward the end of the period. In the South & Islands, the shares of the next 40% and the top 10-1% alternate over time, and the top 1% shows

a gradual increase toward the end of the period, though less pronounced than in the North.

Overall, wealth concentration at the very top increases across all macro areas, but the process is stronger and more persistent in the North.

Figure 3.1: Wealth shares per macro area



Note: Wealth shares are based on weighted net wealth and expressed as proportions of total household wealth.

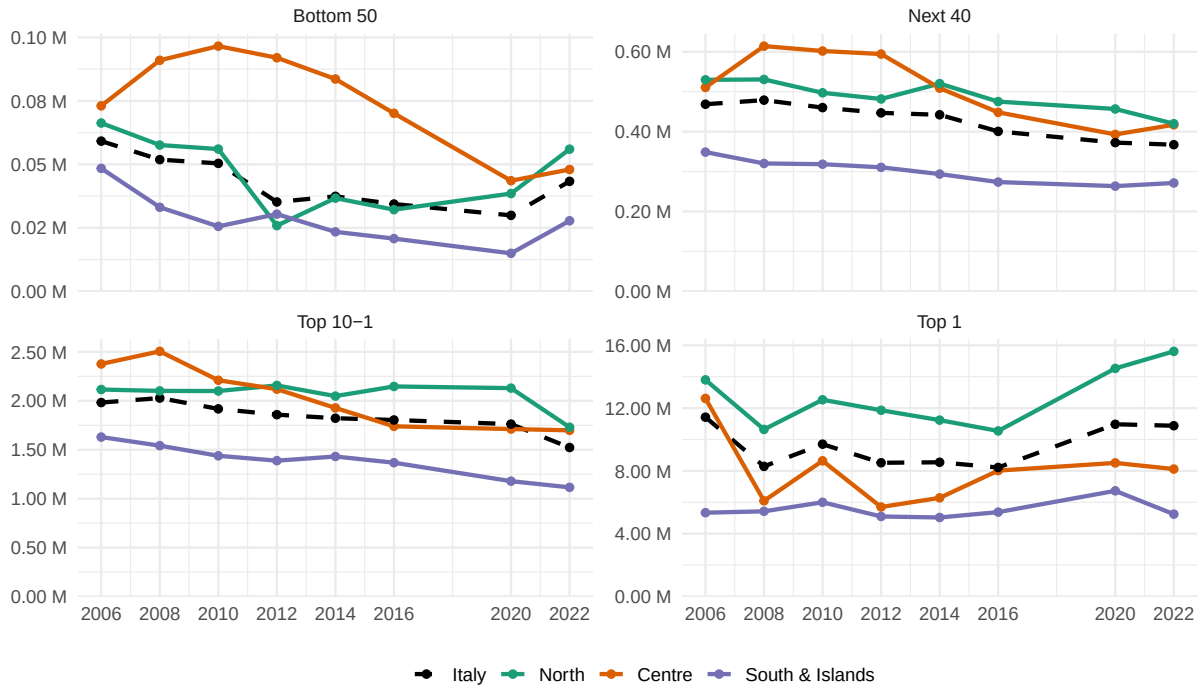
To complement the analysis of wealth shares, examining average wealth by wealth group clarifies whether rising concentration reflects absolute gains at the top, losses elsewhere, or a combination of both.

Figure 3.2 shows that for the bottom 50%, the Centre follows a divergent trajectory from the other macro areas, displaying an increase in average wealth in the early part of the period while the rest of the country experiences decline. However, this pattern reverses over time, as the Centre undergoes a strong contraction after 2010 and is eventually overtaken by the North by the end of the period. The South & Islands remain persistently at lower levels throughout most of the period, with average wealth

remaining around €25,000. Across all macro areas, the upper middle segments (the next 40% and the top 10-1%) exhibits a clear downward trend in average wealth, confirming a progressive erosion of the middle class. The most pronounced divergence, however, concerns the top 1%. In the North, average wealth accelerates sharply after 2016, reaching nearly €16 million by 2022. In the Centre, average wealth at the very top remains comparatively flatter after initial fluctuations, while in the South & Islands it stays consistently below the levels observed elsewhere.

The underlying dynamics are therefore differentiated across macro area. In the North, rising concentration is driven both by gains at the very top and by losses in the middle of the distribution. In the Centre and in the South & Islands, instead, concentration appears to stem primarily from declining wealth among the middle and lower segments, rather than from strong expansion at the top.

Figure 3.2: Average household net wealth by wealth group and macro area



Note: Weighted average household net wealth by wealth group and macro area, in 2022 constant euros. Values are expressed in millions of euros.

Moreover, the widely used Gini coefficient (together with the Lorenz curves), re-

ported in Appendix B.2, broadly confirms the patterns emerging from the previous analyses. The Centre follows a partially distinct trajectory, with an initial phase of declining inequality driven by a temporary strengthening of the lower segment of the distribution. However, this compression is not persistent, and in the later years the Centre converges toward the same upward trend observed in the other macro areas. Overall, the Gini values fluctuate within a range between 0.64 and 0.78, closely matching prior evidence (Acciari et al., 2024). Unsurprisingly, these levels are substantially higher than typical Gini estimates for income,[2] confirming that wealth is structurally more concentrated than income.

3.1.3 Portfolio Composition

Decomposing household portfolios into asset and liability components allows us to examine the type of assets through which wealth evolves over time. In what follows, portfolio shares are computed at the group level as weighted aggregates, with both assets and liabilities normalized by gross assets. Leveraging the detailed micro-level information provided by the SHIW, Figure 3.3 shows portfolio composition across wealth groups, macro areas, and for the years 2006 and 2022. Figure 3.4 further illustrates the change in composition between these two years, highlighting how portfolio allocation has shifted over time.

Focusing first on the lower segment of the wealth distribution, the bottom 50% displays slightly different portfolio structures across macro areas. In the Centre and in the South & Islands, *Dwellings* account for the overwhelming majority of gross assets, indicating that housing is essentially the only form of wealth held by poorer households in these areas. By contrast, in the North, at the beginning of the period, *Dwellings* are relatively less dominant, and even among the poorest households there is a small

[2]World Bank data report an average Gini index for disposable income of approximately 0.35 in Italy over the period 2006–2022.

but non-negligible presence of financial instruments, such as *Bonds* and *Public Equity*. This suggests greater participation in financial markets, even at the bottom of the distribution. However, over time, a common pattern emerges: between 2006 and 2022, the share of *Dwellings* increases across all macro areas, while the already limited share of financial instruments declines. At the same time, liabilities represent a substantial fraction of gross assets for the bottom 50%, often exceeding half of total assets. This leverage has expanded over time, particularly in the Centre and in the South & Islands, reflecting the reliance on mortgages and consumer debt and increasing the vulnerability of less wealthy households to adverse economic shocks.

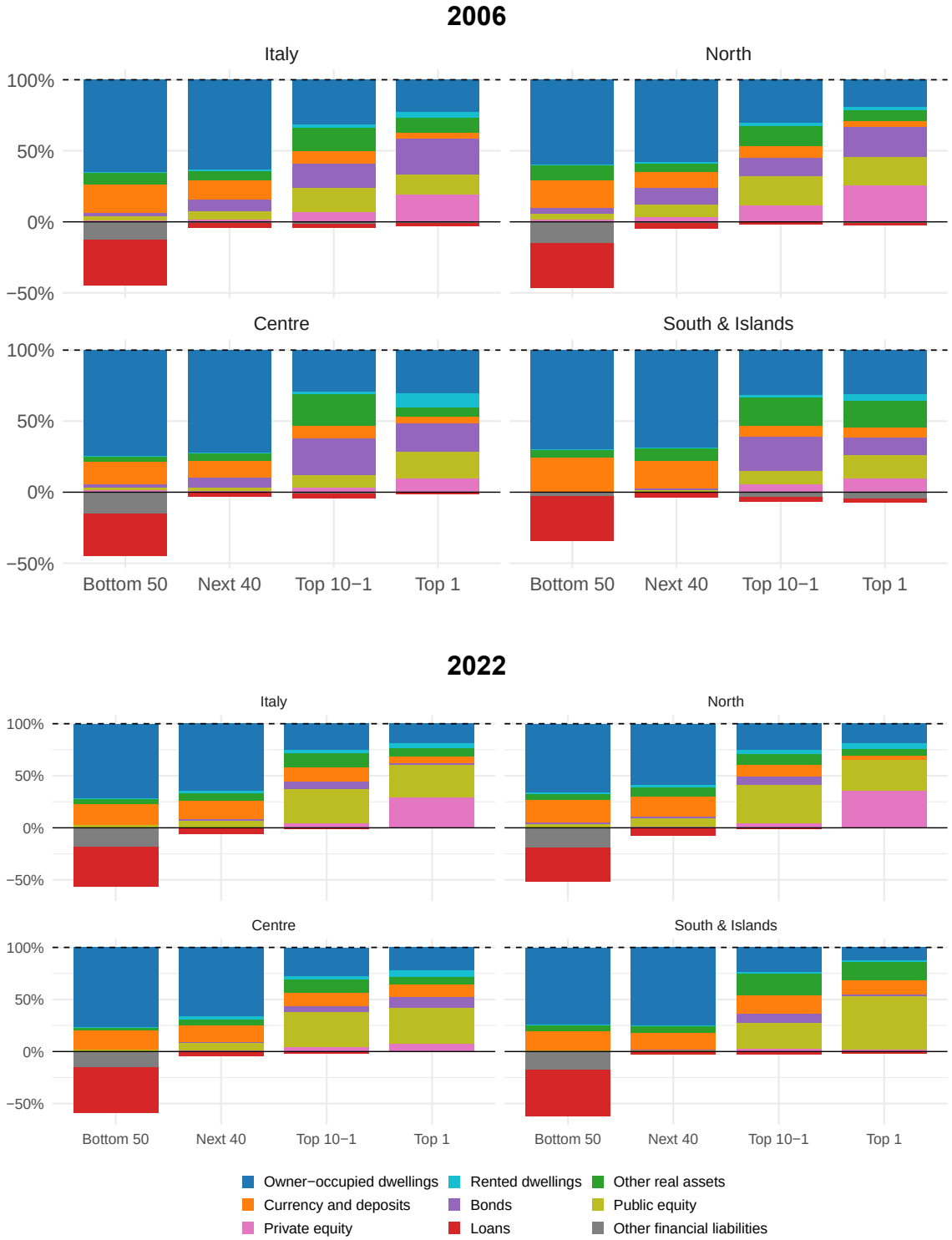
Moving up the distribution, the next 40% exhibits a portfolio structure broadly similar to that of the bottom half. *Dwellings* remain the dominant asset category across all macro areas, while financial instruments are more relevant but still play a secondary role. The main distinction relative to the bottom 50% lies in the significantly lower level of liabilities, which points to a more stable balance-sheet position. Overall, the portfolio composition of this middle segment appears relatively stable over time across all macro areas.

Differences across macro areas become more pronounced in the top decile. In the North, financial instruments already represent a substantial component of portfolios in 2006, particularly for the top 1%, where they exceed 65% of gross assets, and their importance increases further by 2022. In the Centre and in the South & Islands, by contrast, *Dwellings* and *Other real assets* remain relatively more important for top households in the earlier period. The share of financial instruments rises significantly only by 2022, largely through a reallocation from real assets, leading top portfolios to increasingly resemble those observed in the North. Focusing on the internal composition of financial wealth, a common trajectory across all macro areas is the reallocation from *Bonds* toward riskier instruments between 2006 and 2022. A key distinction, however, concerns the type of risky assets held. In the North, a significant share of financial

wealth held by the top 1% is represented by *Private equity*, whereas in the Centre and in the South & Islands portfolios are more heavily concentrated in *Public equity*. Since *Private equity* displays higher average returns over the period, this compositional difference further strengthens the return advantage of top wealth groups in the North relative to the rest of the country.

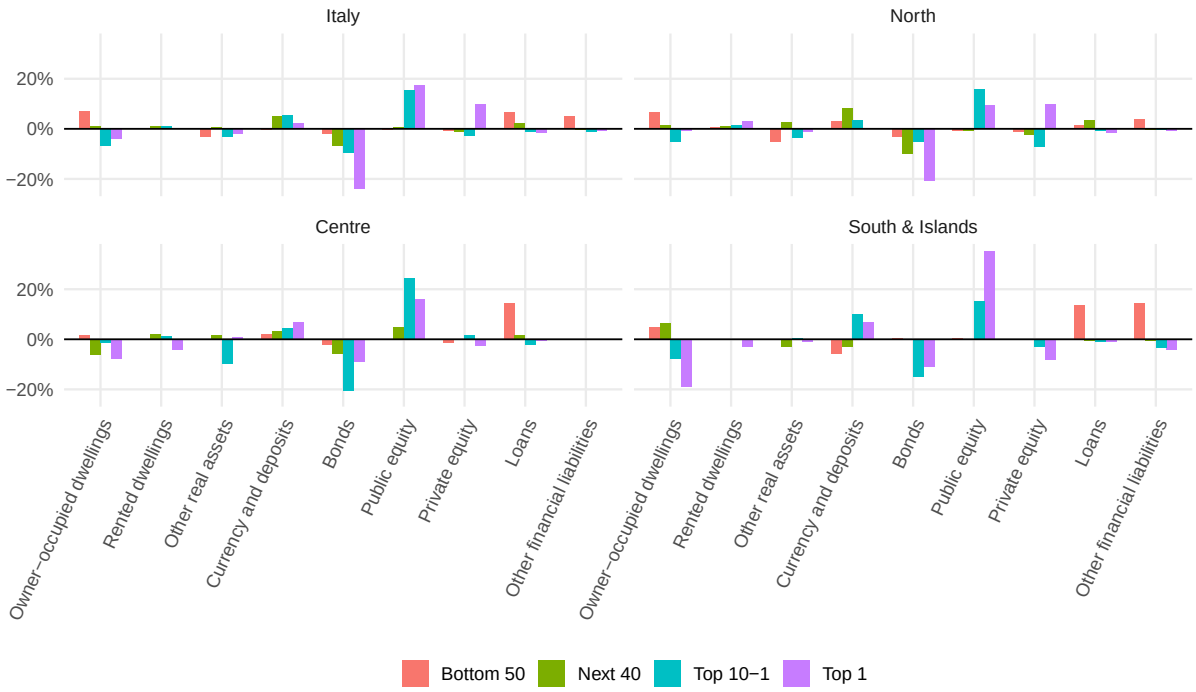
In general, these patterns highlight that differences in where wealth is held are at least as important as differences in how much wealth is held. The composition gap between the top and the rest of the distribution, combined with the relatively higher leverage at the bottom, creates structurally different exposures to asset price dynamics. Moreover, changes in portfolio composition over time appear less pronounced in the North across all wealth groups, suggesting a relatively more stable allocation compared to the other macro areas. Finally, territorial heterogeneity also emerges within financial instruments among top groups: in the North, they tend to hold a larger share of higher risk and higher yield instruments, reinforcing their potential return advantage.

Figure 3.3: Portfolio composition by wealth group and macro area



Note: Bars report the portfolio composition of household net wealth by wealth group and macro area. Asset components (dwellings, other real assets, currency and deposits, and other financial instruments) are expressed as shares of gross assets and sum to 100% within each group. Liability components (loans and other financial liabilities) are expressed as shares of gross assets and are shown as negative values.

Figure 3.4: Change in portfolio composition by wealth group and macro area (2006–2022)



Note: Bars represent the change in the share of each asset or liability in total household wealth between 2006 and 2022. Positive values indicate an increase in the relative importance of the asset, while negative values indicate a decline.

3.2 Returns and Counterfactual Wealth Dynamics

3.2.1 Asset Returns and Portfolio Exposure

After documenting the differences in the composition of household portfolios across wealth groups and macro areas, the analysis turns to the implications of holding different assets in terms of returns. Portfolio heterogeneity could be a relevant channel of wealth inequality only if asset classes exhibit distinct return patterns over time. Appendix B.3 confirm that this is indeed the case, reporting annual real returns for the main asset categories, distinguishing between real assets by macro area and financial assets at the national level.

For real assets, both *Dwellings* and *Other real assets* experienced a prolonged phase of weak performance across all macro areas. Capital gains are negative for a substantial portion of the period, particularly between 2010 and 2015, with only limited and temporary recoveries. The Centre records the most pronounced price reductions, with annual capital losses exceeding 9% in 2013. As a result, households heavily exposed to housing wealth faced persistent downward pressure on asset values. Rental yields, by contrast, provide a relatively stable and positive component of return. However, this income component contributes to greater differentiation across wealth groups, as rental income is more prevalent among wealthier households that own additional properties beyond their primary residence.

For financial assets, *Deposits* and *Bonds* are characterized by low volatility and persistently modest returns. Their contribution to wealth accumulation is relatively stable and, in some periods, becomes negative in real terms. Riskier assets exhibit a different profile. *Public equity* is the most volatile component, with sharp downturns during crisis episodes, such as the 2008 financial crisis when returns approached −50%, followed by strong recoveries in subsequent years, with peaks exceeding 30%. *Private equity*, while displaying lower volatility than listed equity, records persistently positive and comparatively high returns, averaging around 10% per year after 2012.

Given this pronounced heterogeneity in asset performance, portfolio composition emerges as a key determinant of wealth dynamics. Table 3.1 translates asset-specific returns into average portfolio returns by wealth group and macro area, computing weighted averages of asset returns using fixed portfolio shares. The evidence is clear: the Top 1% systematically earns the highest average returns on gross wealth across all macro areas, with the North standing out as the region where these returns are the highest. When 2006 portfolio shares are held fixed, the Top 1% in the North records an average annual real return of 1.89% on gross wealth, while the Centre and the South & Islands display lower values, reflecting portfolios that remain more concentrated in housing.

At the opposite end of the distribution, the Bottom 50% records negative average returns in every macro area. A similar, though less pronounced, pattern emerges for the Next 40%, indicating that even the middle segment of the distribution experienced negative average returns over the period. This largely reflects their strong exposure to real assets during years of declining housing prices. As a robustness check, Panel B reports results obtained by fixing 2022 portfolio shares. The conclusions remain qualitatively unchanged. However, the return differential between top and bottom groups becomes even more pronounced, highlighting the effect of the observed reallocation toward higher-yielding financial assets in the more recent portfolio structure.

This evidence shows that, also in Italy, the average return on wealth increases with household wealth, in line with the findings of Bach et al. (2020), Fagereng et al. (2020) and Xavier (2020) for Sweden, Norway, and the United States, respectively. This pattern suggests that differences in portfolio structure, combined with heterogeneous asset returns, mechanically generate substantial dispersion in average wealth growth across groups. The next subsection formalizes this intuition through a counterfactual simulation.

Table 3.1: Average annual real portfolio returns on gross wealth by macro area and wealth group (%, 2006–2022)

Panel A: 2006 portfolio weights

Area	Bottom 50	Next 40	Top 10-1	Top 1
Italy	-0.93 (2.28)	-0.70 (2.37)	0.48 (3.79)	1.27 (3.27)
North	-0.63 (2.24)	-0.32 (2.64)	0.95 (4.52)	1.89 (4.53)
Centre	-1.51 (2.86)	-1.45 (2.79)	-0.09 (2.23)	0.49 (3.99)
South & Islands	-0.96 (2.19)	-0.84 (2.20)	0.30 (2.34)	0.78 (3.65)

Panel B: 2022 portfolio weights

Area	Bottom 50	Next 40	Top 10-1	Top 1
Italy	-1.10 (2.41)	-0.81 (2.50)	0.79 (6.73)	2.37 (6.83)
North	-0.87 (2.32)	-0.53 (2.65)	0.99 (7.54)	2.76 (6.60)
Centre	-1.66 (2.93)	-1.21 (2.91)	0.58 (6.76)	0.87 (7.07)
South & Islands	-0.96 (2.26)	-0.93 (2.27)	0.55 (5.20)	1.35 (10.45)

Note: Values report mean annual real portfolio returns over 2006–2022; standard deviations are reported in parentheses. Portfolio returns are computed as weighted averages of asset-specific returns using fixed portfolio shares of gross wealth (2006 and 2022 respectively).

3.2.2 Counterfactual Simulation

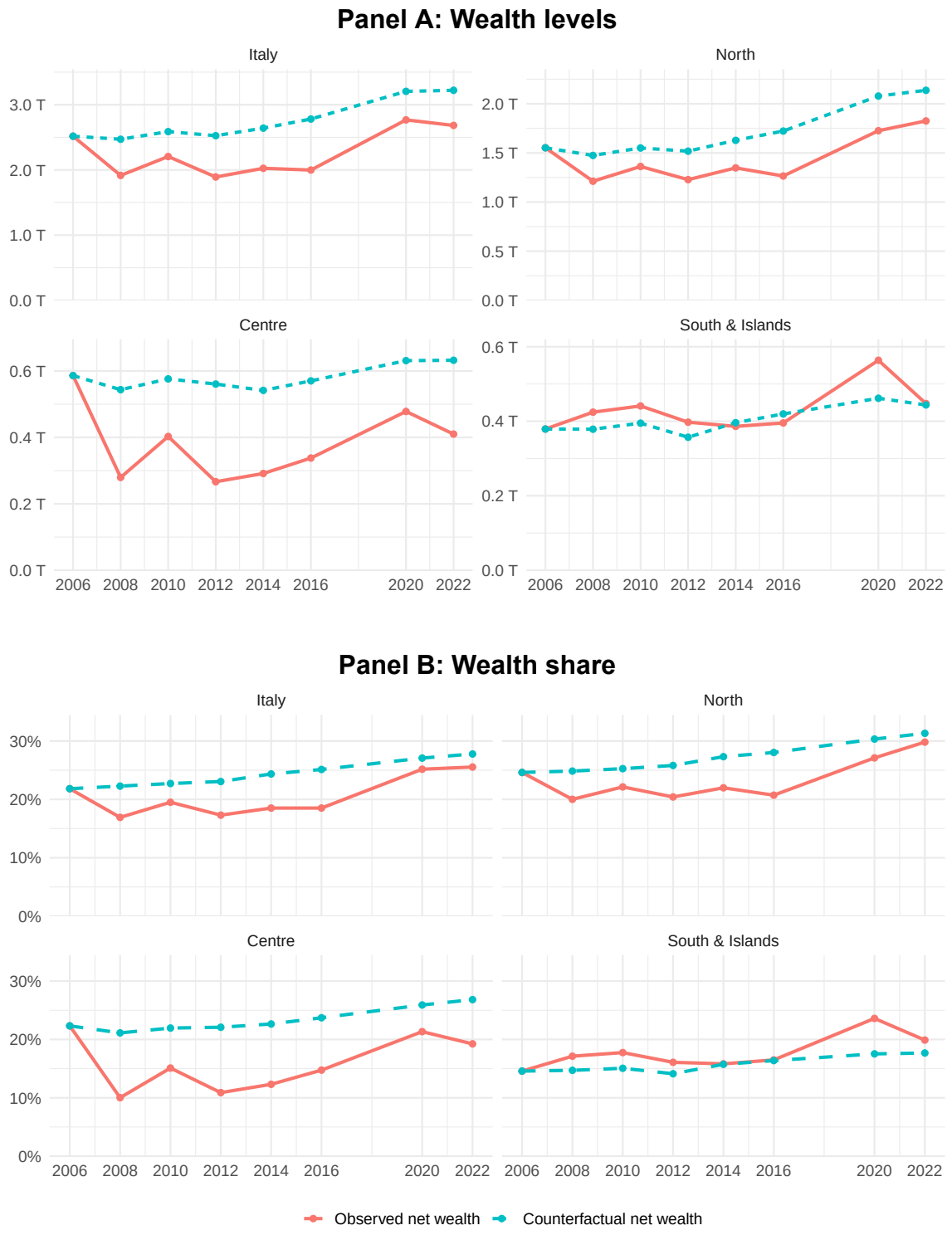
Building on the methodological framework outlined in section 2.4 and on the empirical evidence presented above, this subsection reports the results of the counterfactual simulation in which asset-specific returns are applied to a fixed portfolio composition. The objective is to assess the extent to which the observed evolution of wealth at the top of the distribution can be explained by returns alone.

Figure 3.5 compares observed and counterfactual net wealth levels for the top 1% across macro areas, fixing the 2006 portfolio structure. The counterfactual path, represented by the dotted blue line, closely tracks observed dynamics, particularly in the North and in Italy as a whole. In both cases, although the counterfactual level tends to overestimate wealth in absolute terms, the two series move largely in parallel, displaying similar turning points and growth phases.

In the Centre, the simulation overestimates wealth in the initial years, failing to capture the sharp contraction observed between 2006 and 2008. Nevertheless, the overall trend remains broadly aligned, especially in the second half of the sample. In the South & Islands, the relationship appears more irregular: the counterfactual series alternates between periods of underestimation and overestimation, suggesting that mechanisms other than returns may play a more relevant role in shaping top wealth dynamics in this macro area.

Linking these results to wealth concentration, Figure 3.5 also displays the evolution of the top 1% wealth share under the same fixed portfolio assumption. Although the counterfactual shares are smoother and fail to reproduce the short-run fluctuations observed in the data, they remain consistent with the overall trend. This pattern is particularly evident in the North, where by 2022 the counterfactual and observed shares almost overlap at around 30% of total wealth.

Figure 3.5: Observed vs. counterfactual net wealth of the Top 1% under a fixed 2006 portfolio (2006–2022)



Note: The figure compares observed net wealth to a counterfactual path obtained by applying asset-specific real returns to the 2006 portfolio composition, holding portfolio shares constant over time. Wealth is expressed in constant 2022 euros. The wealth share is computed as the Top 1%'s net wealth relative to total net wealth in each macro area.

Appendix Figures B.4 and B.5 provide complementary evidence that further supports the baseline results. The first figure replicates the simulation for the bottom 50%. Across all macro areas, the overall trend of observed wealth is broadly respected by the counterfactual series, although the degree of alignment varies. In particular, the South & Islands display the closest correspondence between counterfactual and observed dynamics, both in wealth levels and in wealth shares. This suggests that in this macro area asset returns account for a substantial share of the observed evolution specifically for the bottom 50%.

The second appendix figure reports the robustness check based on the 2022 fixed portfolio. The results broadly confirm those obtained under the 2006 fixed portfolio assumption. In several cases, the magnitude of the dynamics appears even more closely aligned, with a smaller gap between observed and counterfactual wealth levels, particularly in the North and in the Centre. Overall, the similarity of the findings across alternative portfolio benchmarks suggests that the explanatory role of returns is not driven by the specific choice of the reference portfolio, but remains robust to different fixed portfolio assumptions.

3.2.3 Regression Results

To quantify the extent to which asset returns alone explain the evolution of wealth, a reduced-form regression is estimated linking observed and counterfactual dynamics. Specifically, for each macro area a and wealth group g, I estimate the following specification:

$$\Delta \log NW^{\text{obs}}_{a,g,t} = \alpha_{a,g} + \beta_{a,g} \, \Delta \log NW^{\text{syn}}_{a,g,t} + \varepsilon_{a,g,t}, \tag{3.1}$$

where $\Delta \log NW^{\text{obs}}_{a,g,t}$ denotes the observed log-growth rate of net wealth between $t-1$ and t, and $\Delta \log NW^{\text{syn}}_{a,g,t}$ is the corresponding log-growth rate generated by the

counterfactual simulation that applies asset-specific returns. The parameter $\beta_{a,g}$ measures the elasticity of observed wealth growth with respect to counterfactual wealth growth. The error term $\varepsilon_{a,g,t}$ captures residual variation not explained by asset returns.

The coefficient of determination R^2 provides a summary measure of explanatory power, indicating the fraction of the variance in observed wealth growth that is accounted for by the simulated returns-only process over the full sample period. A higher R^2 therefore implies that differences in asset returns play a quantitatively important role in shaping the evolution of wealth for the corresponding macro area and wealth group.

The regression results, shown in Table 3.2 confirm and formalize the patterns previously observed in the graphs. Panel A, which focuses on net wealth levels, shows that the counterfactual dynamics explain a large share of the observed variation in top wealth, particularly in Italy and in the North. Under the 2006 fixed portfolio specification, the coefficient of determination R^2 reaches 0.86 for Italy and 0.85 for the North. This implies that approximately 85–86% of the variation in observed log wealth growth of the top 1% can be accounted for by the synthetic wealth dynamics driven solely by asset returns.

The Centre also displays substantial explanatory power, with an R^2 of 0.74 under the 2006 portfolio, and even higher values under alternative portfolio specifications. By contrast, the South & Islands show a much weaker fit, with an R^2 of 0.37 under the baseline specification and even lower when the 2022 portfolio is used. This suggests that in the South & Islands additional mechanisms beyond asset returns play a more prominent role in shaping the evolution of top wealth.

Robustness checks based on the 2022 portfolio and on the average 2006–2022 portfolio confirm the central findings. In most of the macro areas, explanatory power increases further, reaching values above 0.90, particularly for the Centre under the 2022 fixed portfolio. This indicates that the returns-only mechanism remains important across alternative portfolio calibrations.

Panel B turns to the evolution of the top 1% wealth share. As expected, the explanatory power is lower than in the levels regressions, since shares incorporate the dynamics of the entire distribution. Nevertheless, the results remain economically meaningful. For instance, in the North, asset returns alone account for more than half of the observed variation in the top 1% share under the 2006 portfolio, with Italy showing a similar magnitude. The Centre exhibits weaker explanatory power, whereas the South & Islands perform relatively better than in the levels specification.

The increase in R^2 under the 2022 and average portfolio calibrations suggests that more recent portfolio structures better capture the mechanisms underlying changes in concentration.

Table 3.2: Returns-driven wealth dynamics under alternative fixed portfolios: Top 1%

Panel A: Top 1% net wealth (log-growth regression)

	2006 portfolio		2022 portfolio		Average portfolio	
Macro area	β	R^2	β	R^2	β	R^2
Italy	3.20	0.86	1.73	0.92	2.31	0.92
North	2.17	0.85	1.52	0.87	1.80	0.87
Centre	5.69	0.74	3.99	0.93	4.87	0.88
South & Islands	1.50	0.37	0.34	0.09	0.70	0.18

Panel B: Top 1% wealth share (log-growth regression)

	2006 portfolio		2022 portfolio		Average portfolio	
Macro area	β	R^2	β	R^2	β	R^2
Italy	1.10	0.52	0.85	0.67	0.96	0.61
North	1.04	0.56	0.79	0.64	0.90	0.61
Centre	1.44	0.38	1.40	0.39	1.43	0.38
South & Islands	1.62	0.59	0.96	0.45	1.22	0.52

Note: Each cell reports the slope coefficient β and the coefficient of determination R^2 from regressions of observed log-growth on counterfactual log-growth generated under alternative fixed portfolio structures. Panel A refers to net wealth levels; Panel B refers to the wealth share.

Appendix B.2 provides additional evidence by decomposing the previous analysis into real and financial assets. Specifically, the log growth of observed real (financial) assets is regressed on the corresponding log growth of synthetic real (financial) assets

generated by the returns-only simulation. The results clearly show that financial assets display substantially higher R^2 values than real assets across all macro areas. This evidence suggests that the returns-only mechanism is especially effective in accounting for the dynamics of the financial component of wealth, while real asset evolution appears to be influenced by additional factors.

This additional evidence reinforces the main conclusions of the section. First, average portfolio returns systematically increase when moving up the wealth distribution across all macro areas. Second, a large share of the observed evolution of top 1% wealth can be attributed to differential exposure to asset returns, especially in the North and in Italy as a whole. Third, this mechanism is driven primarily by financial assets, whose returns account for a substantial share of the observed dynamics.

Chapter 4: Limitations and Further Research

The empirical results presented in the previous chapter should be interpreted in light of potential data and methodological constraints. This chapter discusses these limitations by distinguishing between those related to the construction of wealth distributions, and those arising from the counterfactual simulation. It then highlights possible directions through which future research could refine the analysis of household wealth inequality in Italy.

4.1 Limits in the Construction of Wealth Distributions

A first set of limitations relates to the intrinsic constraints of the SHIW. As widely discussed, survey data suffer from non-response, under-reporting, and limited coverage of the upper tail of the distribution. Although this work attempts to mitigate all these issues, corrections inevitably rely on additional assumptions and may introduce uncertainty. Moreover, the coverage of the SHIW is itself incomplete. Because the survey is drawn from the *liste anagrafiche comunali* (municipal population registers), it underrepresents immigrant populations, especially irregular migrants, and excludes individuals living in institutions such as prisons, military barracks, retirement homes, or convents. As noted by Baffigi et al. (2016), these omissions may affect measured inequality, since these groups are more likely to belong to the lower end of the distribution. In principle, a survey design that fully captures the entire population would reduce the need for ex post adjustments and improve the reliability of future research.

Additional limitations arise from conceptual discrepancies between microdata and macroeconomic sources. For example, National Accounts include severance pay (TFR) within insurance technical reserves, an item not captured in the SHIW, and adopt market-

value and accrual accounting conventions for life insurance and pension fund components. Further differences arise from valuation criteria. National Accounts record most financial instruments at current market value and on an accrual basis, whereas survey respondents tend to report book or purchase values using cash accounting. As noted by Bonci et al. (2005), these divergent conventions particularly affect securities and interest-bearing assets. Finally, the micro–macro reconciliation itself relies on simplifying assumptions. The proportional adjustment method distributes the gap uniformly across households within each asset category. While alternative approaches exist, they also require strong assumptions (D'Alessio and Neri, 2015). Future research could refine reconciliation methods through improved data integration and more sophisticated alignment procedures.

The use of Forbes billionaire data to inform the upper tail of the wealth distribution also introduces potential limits. Forbes valuations do not constitute an official statistical source and rely on heterogeneous and partly unverifiable information. Additionally, the Forbes list reports individual net worth, whereas both the SHIW and the National and Regional Accounts adopt a household definition of wealth. In principle, aligning the two concepts would require adjusting individual valuations to the appropriate household unit. The literature typically consider this mismatch to be of limited quantitative relevance relative to the broader measurement uncertainties affecting billionaire wealth (Vermeulen, 2018; Bach et al., 2019). Nonetheless, the reliance on external rich lists reflects the difficulty of capturing the very top of the distribution. Strengthening the oversampling of high-wealth households in Italy would therefore represent a key priority for future survey design, reducing the need for external corrections and improving the direct measurement of top wealth shares.

4.2 Limits of the Counterfactual Returns Approach

The portfolio simulation exercise is also subject to some methodological constraints. First, the analysis relies on proxy measures of asset returns rather than on direct observations of effective gains at the household level. For several asset classes, for instance *Other real assets*, available information is limited and requires indirect approximations based on heterogeneous data sources. For financial assets, additional challenges arise in distinguishing between alternative concepts of return, such as market price changes, total returns, and yields to maturity in the case of bonds. The use of benchmark indices and representative yields therefore introduces another source of potential discrepancies. Moreover, due to data constraints, certain assets had to be grouped into broader categories, thereby simplifying their return profiles. In particular, the distinction between government and corporate bonds was not implemented in the final specification due to data limitations in reconstructing consistent return series for each category over the full sample period. As a result, a common return was applied to the broader bond component, implying that potential differences in performance are not explicitly captured.

More generally, the simulation assumes homogeneous returns within each asset category and macro area. However, recent empirical evidence shows that returns to wealth can be heterogeneous and persistent even within narrowly defined asset classes, with systematically higher returns accruing to wealthier households (Fagereng et al., 2020; Smith et al., 2023). Future research could refine the analysis by constructing more granular and reliable asset-specific return series, particularly for less transparent components such as private business wealth or complex financial instruments.

A further limitation concerns the difficulty of disentangling the effect of returns from other channels of wealth accumulation. In the simulation framework, changes in wealth are attributed to asset-specific returns applied to existing holdings. However, observed wealth dynamics may also reflect net saving behavior, inter vivos transfers, or consump-

tion decisions. It is therefore challenging to fully separate movements driven by capital gains and return differentials from those resulting from the addition or withdrawal of wealth units. A richer framework incorporating savings and transfer dynamics would allow a clearer decomposition between return driven accumulation and income based wealth formation.

Finally, while the analysis adopts a macro-area perspective, a more granular regional disaggregation could provide additional insights. Patterns observed at the macro-area level may mask substantial heterogeneity across individual regions. With more detailed data, it would be possible to assess whether return-driven dynamics differ within macro areas and whether spatial wealth concentration exhibits additional layers of divergence.

Conclusions

This work has shown that, although wealth remains a central pillar of household economic security in Italy, its evolution between 2006 and 2022 has been marked by a persistent decline in real terms. This downward trajectory parallels the stagnation observed in other macroeconomic indicators and has not prevented a simultaneous increase in concentration at the top of the distribution.

Wealth inequality has intensified over the period, though with territorial differences. The North stands out as the macro area where concentration at the very top has become most pronounced: the top 1% has expanded significantly both in terms of wealth shares and average wealth levels. Although less sharply, signs of increasing concentration at the top are also visible in the Centre and in the South & Islands. At the same time, a gradual erosion of the middle segment of the distribution emerges across all macro areas. The next 40% and the top 10-1% experience weakening average wealth, signaling a compression of the upper middle class. The South & Islands remain structurally poorer in absolute terms, reinforcing long-standing territorial disparities.

The analysis then turns to the mechanisms underlying these dynamics. Portfolio composition differs systematically across wealth groups and macro areas, and asset classes exhibit markedly heterogeneous return profiles. Wealthier households hold a disproportionately large share of financial assets, including public and private equity, while lower and middle groups remain heavily exposed to housing and display higher liabilities. Moreover, average portfolio returns rise systematically along the wealth distribution across all macro areas, implying that differences in portfolio allocation generate persistent gaps in wealth growth over time. The counterfactual simulations confirm this mechanism quantitatively: for the top 1% in the North, asset returns alone account for approximately 85% of the observed variation in wealth growth under the fixed 2006 port-

folio specification. This indicates that returns-driven dynamics explain the vast majority of fluctuations at the very top in this macro area. More broadly, the explanatory power of the returns-only mechanism is substantial at the national level and in the Centre as well, though weaker in the South & Islands, where other factors appear to play a more relevant role.

Overall, the findings indicate that the interaction between portfolio composition and asset returns constitutes a powerful engine of wealth divergence. In a context of stagnant aggregate wealth, differential exposure to high return assets amplifies inequality dynamics. The regional perspective adopted in this work proves essential for identifying where such mechanisms are most pronounced and for understanding how national inequality trends are shaped by spatial heterogeneity. These insights are particularly relevant for policy design, as they suggest that addressing wealth inequality requires not only redistributive measures but also interventions that influence the structural determinants of returns. Without targeted and territorially differentiated policy interventions, the combined effect of stagnant wealth accumulation and structurally uneven returns could deepen the regional divides and reinforce persistent inequalities.

Appendix A: Supplementary Tables for Chapter 2

Table A.1: Correction factors between National and Regional Accounts by asset or liability and year

Asset / Liability	2006	2008	2010	2012	2014	2016	2020	2022
Dwellings	1.00	1.03	1.01	1.00	1.00	1.00	1.00	1.00
Other real assets	1.09	1.12	1.02	1.00	1.00	1.00	1.00	1.00
Non-financial assets	1.01	1.05	1.01	1.00	1.00	1.00	1.00	1.00
Currency and deposits	1.03	1.03	1.02	1.00	1.00	1.00	1.00	1.00
Financial instruments (1)	1.07	1.03	1.01	1.02	1.00	1.00	1.00	1.00
Other financial assets	1.05	1.02	0.99	1.00	1.00	1.00	1.00	1.00
Financial assets	1.06	1.03	1.01	1.00	1.00	1.00	1.00	1.00
Loans	1.01	1.01	1.00	1.00	1.00	1.00	1.00	1.00
Other financial liabilities	0.93	1.01	1.00	1.00	1.00	1.00	1.00	1.00
Financial liabilities	0.99	1.01	1.00	1.00	1.00	1.00	1.00	1.00
Net wealth	1.04	1.04	1.01	1.00	1.00	1.00	1.00	1.00

Note: (1) includes debt securities, shares, holdings, mutual funds, and loans to cooperatives. Correction factors are ratios of National Account values over the sum of Regional Accounts values, by asset or liability and year. A factor greater than 1 means that Regional Accounts underestimate the corresponding value.

Table A.2: Residual discrepancies between totals and sum of sub-items by macro area and year (%)

North

Asset / Liability	2006	2008	2010	2012	2014	2016	2020	2022
Non-financial assets	0.06	0.01	0.00	-0.01	0.00	0.00	-0.02	-0.02
Financial assets	-0.24	0.01	0.07	0.53	-0.02	0.00	0.02	0.00
Financial liabilities	0.00	0.01	-0.05	-0.05	0.00	0.02	0.03	-0.08
Net wealth	-0.82	0.10	0.02	0.00	0.00	0.01	0.00	-0.01

Centre

Asset / Liability	2006	2008	2010	2012	2014	2016	2020	2022
Non-financial assets	0.12	0.26	0.03	0.03	-0.01	0.00	0.00	0.05
Financial assets	0.15	0.01	0.10	-0.90	-0.02	0.01	0.01	-0.01
Financial liabilities	-0.20	-0.02	-0.03	0.11	0.04	0.00	0.30	0.08
Net wealth	0.95	-0.10	-0.05	0.00	0.00	-0.01	0.01	0.03

South and Islands

Asset / Liability	2006	2008	2010	2012	2014	2016	2020	2022
Non-financial assets	-0.23	-0.31	-0.04	-0.01	0.00	0.00	0.04	0.00
Financial assets	0.61	-0.03	-0.32	-0.80	0.09	-0.02	-0.09	0.01
Financial liabilities	0.19	-0.01	0.16	0.01	-0.05	-0.03	-0.37	0.12
Net wealth	0.99	-0.12	0.00	0.00	0.00	-0.02	-0.01	0.00

Note: Residuals are computed as the percentage difference between published totals and the sum of sub-items for each aggregate and year.

Table A.3: Comparison between SHIW and Regional Accounts wealth concepts

Variable	Description	SHIW	NA/RA
Net wealth		x	x
Real assets / Non-financial assets		x	x
Dwellings		x	x
Other real assets	Non-residential buildings, land and improvements, plant, machinery, and other fixed assets		x
Businesses	Real assets of household-owned businesses where members work as self-employed	x	
Valuables		x	
Financial assets		x	x
Currency and deposits		x	x
Debt securities, shares, holdings, mutual funds, loans to cooperatives			x
Other financial assets (RA)	Insurance and social security reserves, trade credits, and other accounts receivable		x
Other financial assets (SHIW)	Bonds, mutual funds, shares, shares in Ltd., partnership shares, foreign securities, and loans to cooperatives	x	
Government bonds		x	
Credits to relatives and friends		x	
Financial liabilities		x	x
Loans		x	x
Other financial liabilities	Insurance, pension, and standardized guarantee schemes and other accounts payable		x
Commercial debts		x	
Debts towards other households		x	

Source: Methodological documentation of the Survey of Household Wealth and Income (SHIW) and Regional Accounts.

Table A.4: Share of NPISH in National Accounts (%)

Asset / Liability	2006	2008	2010	2012	2014	2016	2020	2022
Non-financial assets	5.89	5.43	5.37	5.40	5.63	5.71	5.91	5.99
Financial assets	1.22	1.32	1.26	1.27	1.23	1.21	1.23	1.24
Financial liabilities	1.21	1.13	1.21	1.28	1.22	1.12	0.92	0.89

Note: Shares of Non-Profit Institutions Serving Households (NPISH) over the National and Regional Accounts values of the Households + NPISH sector.

Table A.5: Pareto tail parameters by macro area and selected years

North

Statistic	2006	2010	2016	2022
w_{min} (euros)	1,924,381.6	1,456,451.8	877,019.8	934,981.0
α	1.81	1.85	2.02	1.50
Tail observations	229	228	265	275
Tail population (weighted)	110,183.66	288,732.97	606,018.15	614,933.80

Centre

Statistic	2006	2010	2016	2022
w_{min} (euros)	1,542,590.5	1,154,276.1	1,977,648.8	740,200.0
α	1.61	2.05	2.70	2.05
Tail observations	256	236	227	247
Tail population (weighted)	114,565.69	228,978.08	52,985.25	250,842.29

South and Islands

Statistic	2006	2010	2016	2022
w_{min} (euros)	642,746.0	629,605.1	463,732.9	450,167.0
α	2.21	1.95	2.36	1.99
Tail observations	400	400	400	400
Tail population (weighted)	372,993.37	374,464.73	419,409.16	405,423.07

Note: w_{min} is the Pareto threshold; α is the estimated Pareto shape parameter. Tail observations count the number of units with wealth above w_{min} in the adjusted micro data; tail population corresponds to their weighted sum.

Table A.6: Top 1% wealth share before and after Pareto tail adjustment (%)

North

Statistic	2006	2008	2010	2012	2014	2016	2020	2022
Top 1% (raw)	12.6%	12.9%	13.5%	13.8%	12.6%	12.0%	15.9%	25.8%
Top 1% (adjusted)	15.0%	14.3%	16.8%	15.3%	15.5%	17.1%	20.1%	25.2%

Centre

Statistic	2006	2008	2010	2012	2014	2016	2020	2022
Top 1% (raw)	13.3%	8.4%	10.7%	8.1%	9.2%	10.5%	13.5%	11.6%
Top 1% (adjusted)	17.9%	7.9%	13.1%	9.5%	10.7%	13.2%	14.3%	13.6%

South and Islands

Statistic	2006	2008	2010	2012	2014	2016	2020	2022
Top 1% (raw)	8.6%	17.3%	11.2%	20.8%	10.2%	12.7%	17.0%	11.5%
Top 1% (adjusted)	12.3%	10.5%	13.7%	12.6%	11.5%	11.2%	15.9%	13.4%

Note: Top 1% shares are computed with survey weights. Raw refers to original SHIW distribution, adjusted refers to Pareto-imputed distributions. Values are expressed as percentages.

Appendix B: Supplementary Tables for Chapter 3

Table B.1: Summary wealth indicators by macro area and year

Italy

Indicator	2006	2008	2010	2012	2014	2016	2020	2022
Total wealth (T)	11.5	11.3	11.3	10.9	10.9	10.8	11.0	10.5
Households (M)	22.6	23.4	23.6	24.3	24.6	25.5	25.4	25.4
Wealth per household (k)	510.2	483.7	478.8	449.6	445.2	422.9	433.5	414.3
Wealth/GDP ratio	5.8	5.7	5.8	5.9	5.9	5.6	6.0	5.3

North

Indicator	2006	2008	2010	2012	2014	2016	2020	2022
Total wealth (T)	6.3	6.1	6.2	6.0	6.1	6.1	6.4	6.1
Households (M)	10.9	11.3	11.4	11.9	11.7	12.1	11.8	12.2
Wealth per household (k)	576.8	536.9	537.4	505.7	526.2	504.1	541.2	500.7
Wealth/GDP ratio	5.8	5.5	5.8	5.9	6.0	5.7	6.2	5.4

Centre

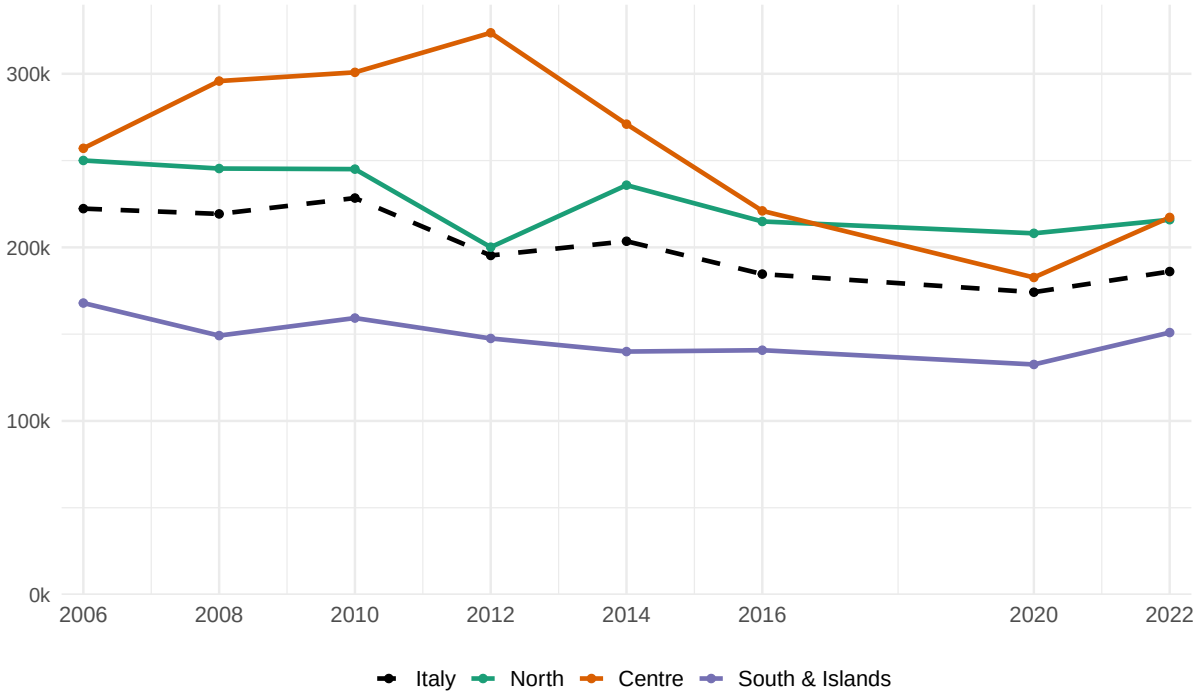
Indicator	2006	2008	2010	2012	2014	2016	2020	2022
Total wealth (T)	2.6	2.8	2.7	2.4	2.4	2.3	2.2	2.1
Households (M)	4.5	4.9	4.6	4.6	5.0	5.2	5.2	5.0
Wealth per household (k)	584.6	574.2	575.5	533.8	477.7	438.1	426.9	424.8
Wealth/GDP ratio	5.9	6.3	6.2	6.0	5.9	5.5	5.7	5.0

South & Islands

Indicator	2006	2008	2010	2012	2014	2016	2020	2022
Total wealth (T)	2.6	2.5	2.5	2.5	2.4	2.4	2.4	2.2
Households (M)	7.2	7.3	7.5	7.8	8.0	8.2	8.4	8.1
Wealth per household (k)	362.3	340.7	330.1	315.3	306.5	293.1	286.0	277.5
Wealth/GDP ratio	5.5	5.3	5.5	5.7	5.9	5.6	5.9	5.1

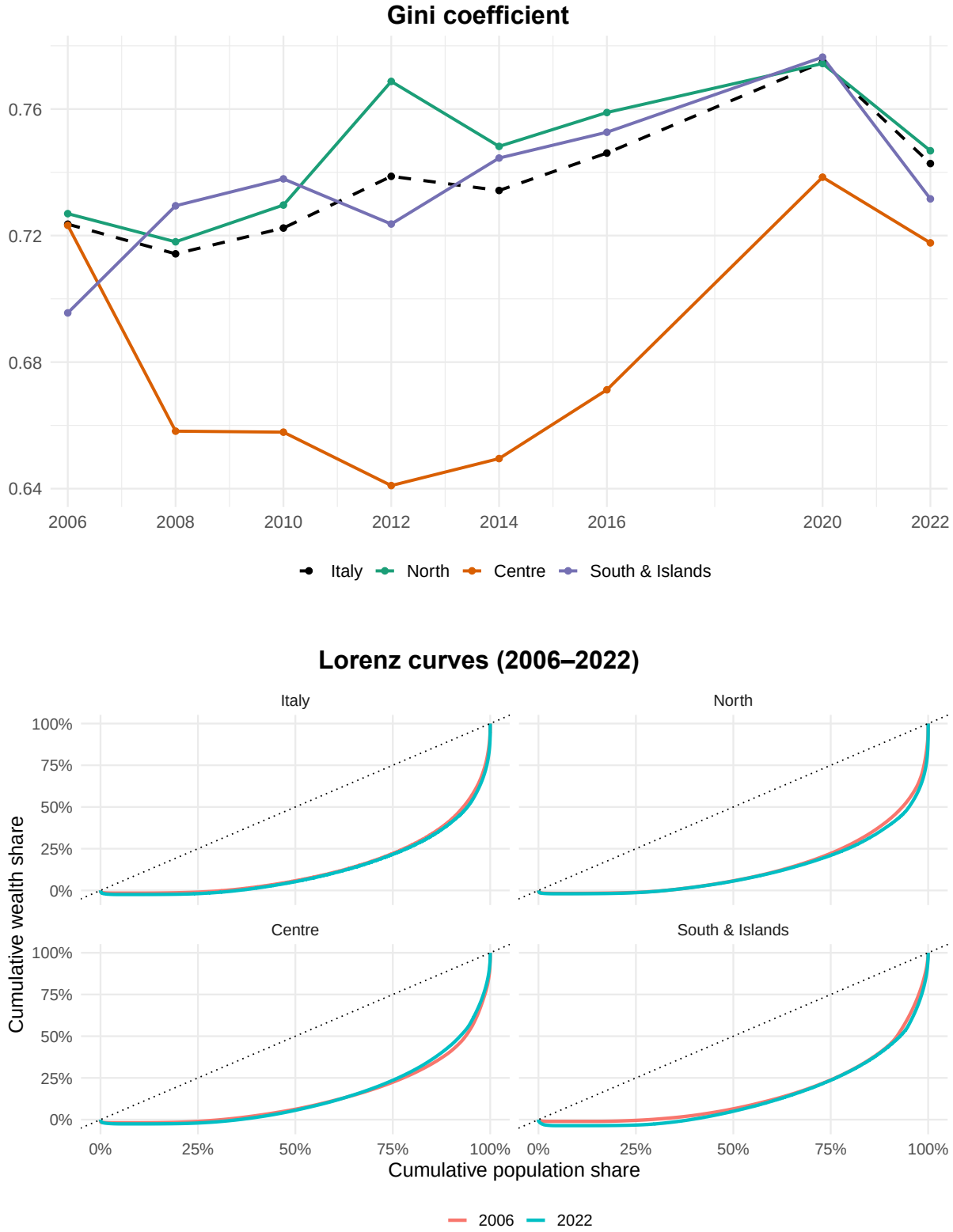
Note: Total wealth is expressed in trillion euros (2022 prices). Household counts are expressed in millions. Wealth per household is shown in thousand euros (2022 prices). Wealth-to-GDP ratio is computed using real GDP for each macro area.

Figure B.1: Median Real Wealth (2022 constant prices) per year and macro areas



Note: Median wealth is calculated as the weighted 50th percentile of household net wealth using SHIW population weights. All figures are expressed in 2022 constant thousand euros.

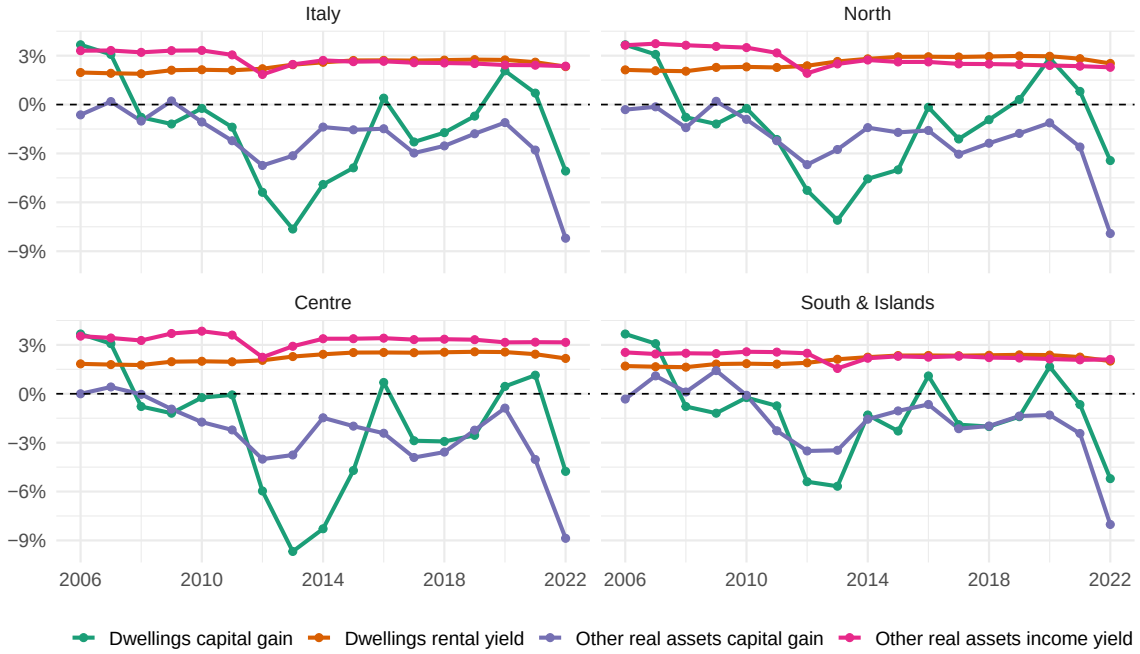
Figure B.2: Wealth inequality indicators by macro area



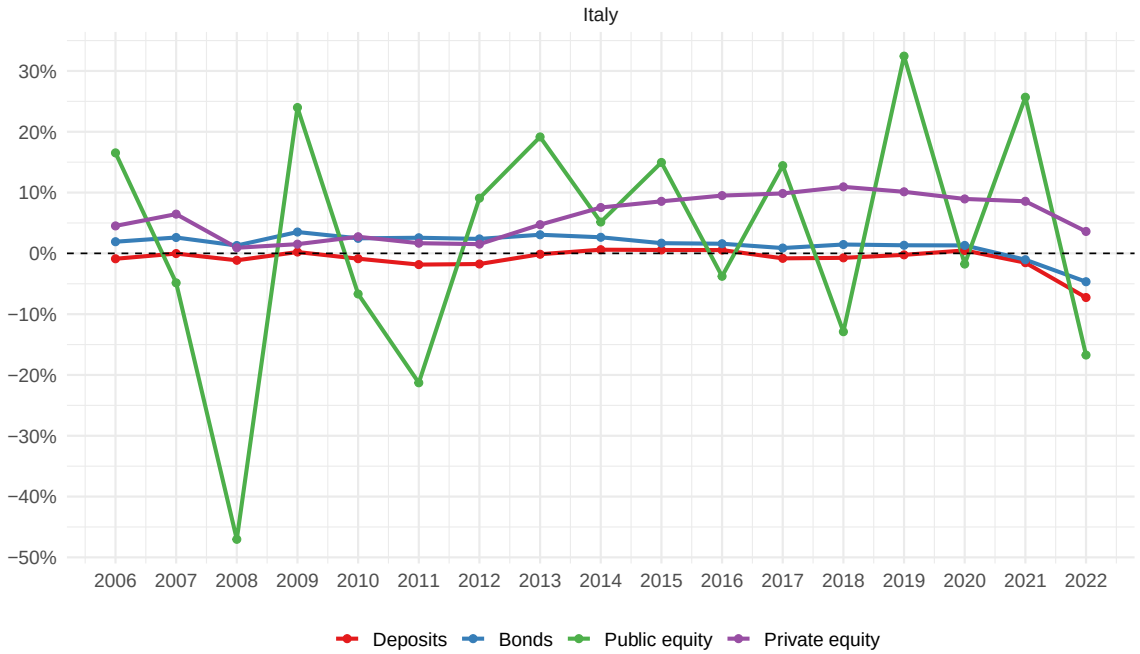
Note: The Gini coefficient measures wealth concentration (0 = perfect equality; 1 = maximum inequality). Lorenz curves depict the cumulative share of wealth held by each share of the population. The farther the curve lies from the 45-degree line, the greater the concentration of wealth. Estimates are computed from weighted household net wealth, expressed in 2022 euros.

Figure B.3: Annual real returns by asset class (2006-2022)

Real assets

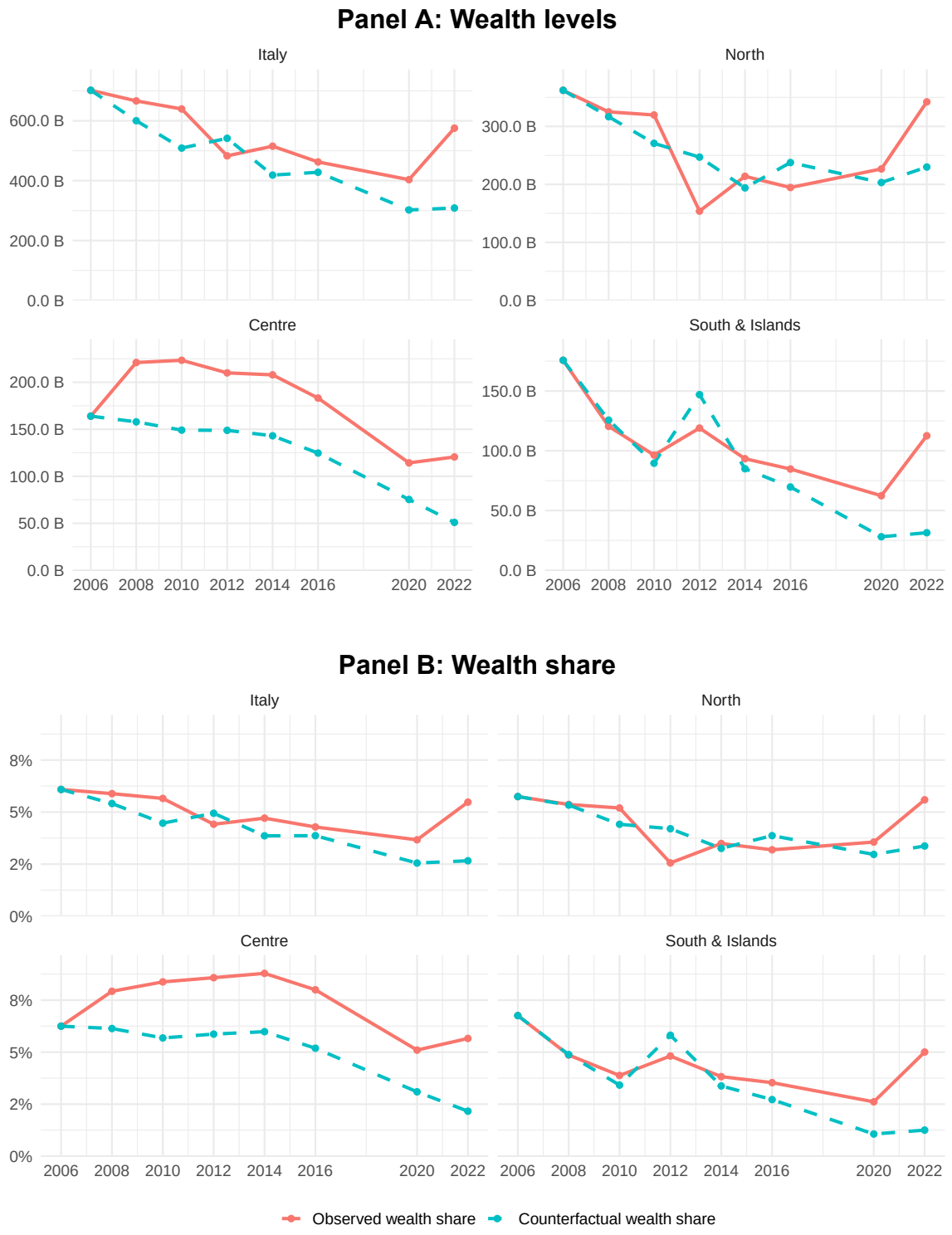


Financial assets



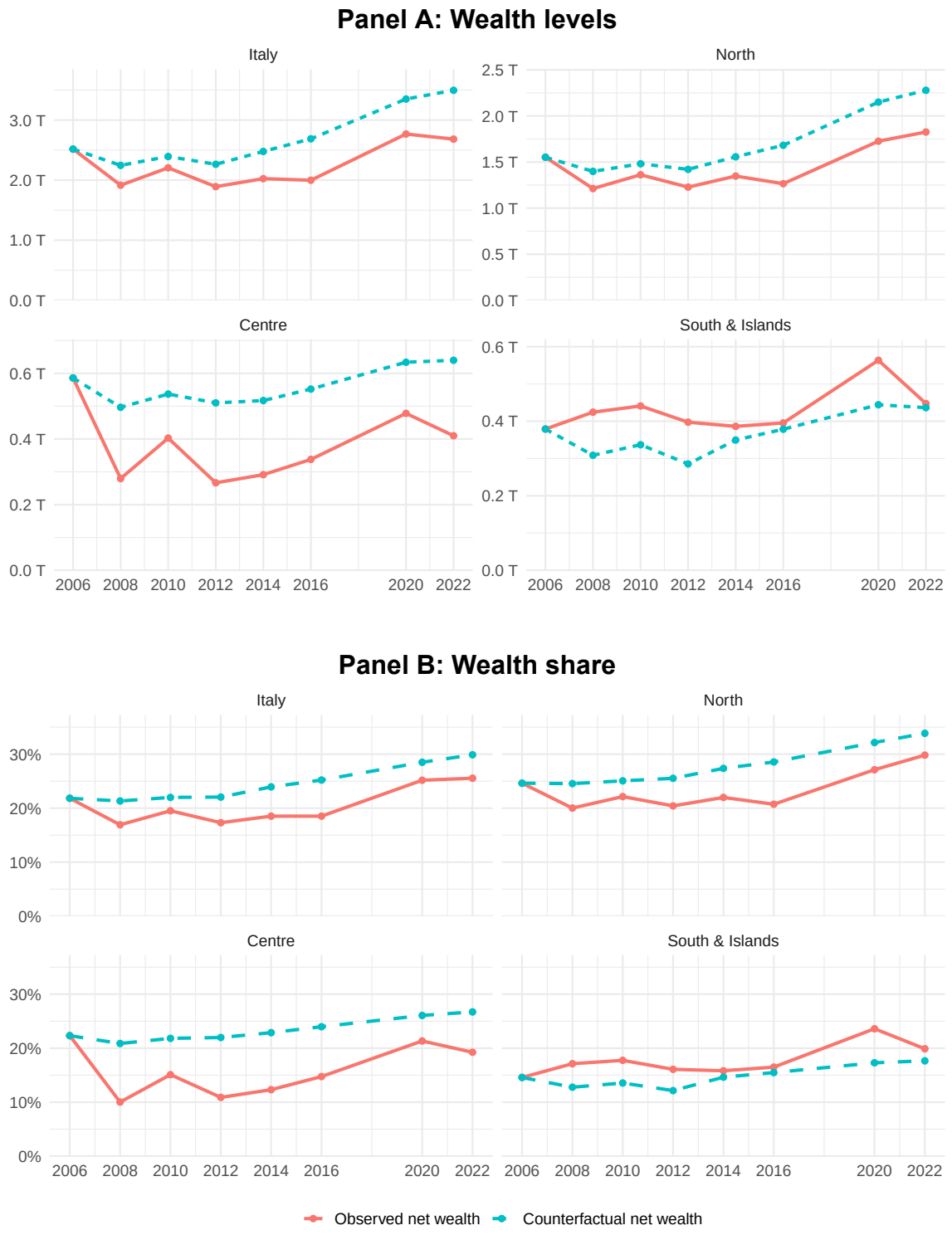
Note: Returns are expressed in real terms.

Figure B.4: Observed vs. counterfactual net wealth of the Bottom 50% under a fixed 2006 portfolio (2006–2022)



Note: The figure compares observed net wealth to a counterfactual path obtained by applying asset-specific real returns to the 2006 portfolio composition, holding portfolio shares constant over time. Wealth is expressed in constant 2022 euros. The wealth share is computed as the Bottom 50%'s net wealth relative to total net wealth in each macro area.

Figure B.5: Observed vs. counterfactual net wealth of the Top 1% under a fixed 2022 portfolio (2006–2022)



Note: The figure compares observed net wealth to a counterfactual path obtained by applying asset-specific real returns to the 2022 portfolio composition, holding portfolio shares constant over time. Wealth is expressed in constant 2022 euros. The wealth share is computed as the Top 1%'s net wealth relative to total net wealth in each macro area.

Table B.2: Real and financial assets returns for Top 1% under 2006 fixed portfolio: regression results

	Real Assets		Financial Assets	
Macro Area	β	R^2	β	R^2
Italy	0.95	0.12	4.78	0.82
North	-0.05	0.00	3.28	0.74
Centre	3.96	0.41	10.11	0.91
South & Islands	1.74	0.12	0.51	0.42

Note: For Real Assets, the dependent variable is the log growth of observed real assets and the independent variable is the log growth of synthetic real assets constructed using time-varying returns. For Financial Assets, the dependent variable is the log growth of observed financial assets and the independent variable is the log growth of synthetic financial assets. β measures the elasticity of observed growth with respect to synthetic growth. R^2 captures the fraction of observed variation explained by the returns within each asset class.

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